

AN INTRODUCTION TO HOMOGENEOUS DYNAMICS

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These lecture notes were written on the occasion of a topics course in dynamics taught at the University of Science and Technology of China from September to December 2026. The aim of the course is to give a rigorous and research-oriented introduction to homogeneous dynamics, with particular emphasis on the space

$$X_2 = \mathrm{SL}_2(\mathbb{R})/\mathrm{SL}_2(\mathbb{Z})$$

and, more generally, on quotients of the special linear groups $G = \mathrm{SL}_d(\mathbb{R})$.

The course is organized around **three** central rigidity and equidistribution phenomena:

- The classification of horocycle orbits and invariant measures on X_2 , culminating in Ratner's theorem in this setting;
- The equidistribution of long closed horocycles in X_2 , proved by combining the geometry of the geodesic and horocycle flows with mixing;
- Random walks on X_2 and the rigidity of stationary measures, in the spirit of the work of Benoist and Quint.

Along the way we develop the basic geometric, ergodic, and arithmetic tools needed to understand these results and their applications. These three themes illustrate complementary mechanisms behind rigidity in homogeneous dynamics—unipotent dynamics, mixing and renormalisation, and random matrix products—and provide an entry point to several directions of current research, including homogeneous equidistribution, Diophantine approximation, measure rigidity, and random walks on homogeneous spaces.

0. WHAT IS HOMOGENEOUS DYNAMICS?

The one-sentence answer to the question in the title is this: one studies the orbits of a subgroup H acting on a quotient $X = G/\Gamma$, where G is a Lie group and Γ a discrete subgroup. An important example is given by $G = \mathrm{SL}_d(\mathbb{R})$, $\Gamma = \mathrm{SL}_d(\mathbb{Z})$, and

$$X_d = \mathrm{SL}_d(\mathbb{R})/\mathrm{SL}_d(\mathbb{Z}) \quad (d \geq 2),$$

which we shall encounter frequently in this course; it can be identified with the *space of unimodular lattices* in $\mathbb{R}^d$ (§2.2). This example is typical: the quotients one studies are not arbitrary spaces, but spaces whose points *are* arithmetic objects — lattices, quadratic forms, tori — and consequently statements about orbits translate into number-theoretic statements. The translation goes both ways, and it is often the case that a question which is intractable as a question about integers becomes tractable as a question about a single orbit.

0.1. A toy model: the two-torus. Before studying the dynamical picture of $\mathrm{SL}_d(\mathbb{R})/\mathrm{SL}_d(\mathbb{Z})$, we treat a simple case in which everything can be proved in a page, and which already has the shape of the general theory. Take

$$G = \mathbb{R}^2, \quad \Gamma = \mathbb{Z}^2, \quad X = G/\Gamma = \mathbb{T}^2,$$

so that X is the two-torus with its Lebesgue probability measure m . Fix $v \in \mathbb{R}^2 \setminus \{0\}$ and consider the subgroup $H = \mathbb{R}v$. We let H act on $\mathbb{T}^2$ by translation,

$$\varphi_t(x) = x + tv \quad (x \in \mathbb{T}^2, t \in \mathbb{R}),$$

so that $H \cdot x = \{\varphi_t(x) : t \in \mathbb{R}\}$. The orbits are the lines of direction v , wrapped around the square. Given a point $x \in \mathbb{T}^2$ and a subgroup H as above, typical questions are:

- Is the orbit $H \cdot x$ closed? Is it periodic? Is it dense in $\mathbb{T}^2$?
- How do these properties depend on x and on H ?
- Which probability measures on $\mathbb{T}^2$ are invariant under the action of H ? Can we classify the ergodic ones?
- Does the orbit $H \cdot x$ *equidistribute* with respect to m (that is, informally, does it spend in each region a proportion of its time equal to the measure of that region)?

The following theorem addresses all of these questions.

Theorem 0.1 (Linear flows on $\mathbb{T}^2$). *Let $v \in \mathbb{R}^2 \setminus \{0\}$.*

- (a) (Rational direction.) *If $v = \lambda w$ with $w \in \mathbb{Z}^2$ primitive and $\lambda \neq 0$, then every orbit is periodic of period $1/|\lambda|$. The orbit closures are the cosets of the circle subgroup $Y = (\mathbb{R}w + \mathbb{Z}^2)/\mathbb{Z}^2$, and the ergodic φ -invariant probability measures are exactly the arclength measures on these circles.*
- (b) (Irrational direction.) *Otherwise every orbit is dense, and in fact equidistributes: for every $f \in C(\mathbb{T}^2)$,*

$$\frac{1}{T} \int_0^T f(x + tv) dt \xrightarrow{T \rightarrow \infty} \int_{\mathbb{T}^2} f dm,$$

uniformly in x . The only ergodic φ -invariant probability measure is m .

In both cases every orbit closure is a coset of a closed connected subgroup of $\mathbb{T}^2$, and every ergodic invariant measure is the Haar measure of such a coset.



FIGURE 1. Orbits of φ_t on $\mathbb{T}^2$, drawn in the unit square with opposite sides identified. Left: $v = (1, 2)$, a rational direction; the orbit closes up after time 1. Right: $v = (1, \varphi)$ with $\varphi = \frac{1+\sqrt{5}}{2}$; the same orbit followed for time 28, on its way to filling the square evenly.

The whole proof is one computation with characters. For $n \in \mathbb{Z}^2$ let

$$e_n: \mathbb{T}^2 \rightarrow \mathbb{C}, \quad e_n(x) = e^{2\pi i \langle n, x \rangle},$$

which is well defined on $\mathbb{T}^2$ because $\langle n, \cdot \rangle$ is integer valued on $\mathbb{Z}^2$. Two standard facts will be used: the linear span of the e_n is dense in $C(\mathbb{T}^2)$ for the uniform norm (Stone–Weierstrass), and a finite Borel measure μ on $\mathbb{T}^2$ is determined by its Fourier

coefficients $\widehat{\mu}(n) = \int e_{-n} d\mu$. The point of the characters is that they turn the flow into multiplication by a scalar:

$$(0.1) \quad e_n(\varphi_t(x)) = e^{2\pi i t \langle n, v \rangle} e_n(x).$$

Lemma 0.2. *There exists $n \in \mathbb{Z}^2 \setminus \{0\}$ with $\langle n, v \rangle = 0$ if and only if v is a real multiple of an integer vector.*

Proof. If $v = \lambda w$ with $w = (w_1, w_2) \in \mathbb{Z}^2 \setminus \{0\}$, take $n = (-w_2, w_1)$. Conversely, $\langle n, v \rangle = 0$ with $n \neq 0$ says that v lies on the line $\mathbb{R}(-n_2, n_1)$, which is spanned by an integer vector. $\square$

Proof sketch of Theorem 0.1. Periods. The stabiliser $\{t \in \mathbb{R} : \varphi_t(x) = x\}$ equals $\{t : tv \in \mathbb{Z}^2\}$ and does not depend on x . If $v = \lambda w$ with w primitive, then $tv \in \mathbb{Z}^2$ if and only if $t\lambda \in \mathbb{Z}$, so the stabiliser is $\lambda^{-1}\mathbb{Z}$ and the orbit is periodic of period $1/|\lambda|$; its image is the coset $x + Y$. If v is not a multiple of an integer vector, then $tv \in \mathbb{Z}^2$ forces $t = 0$ and no orbit is periodic.

Equidistribution in case (b). For $T > 0$ define the *averaging operator*

$$A_T : C(\mathbb{T}^2) \longrightarrow C(\mathbb{T}^2), \quad (A_T f)(x) = \frac{1}{T} \int_0^T f(x + tv) dt,$$

and write $m(f) = \int_{\mathbb{T}^2} f dm$ for the mean of f . That $A_T f$ is again continuous follows from the uniform continuity of f on the compact space $\mathbb{T}^2$. In fact, we shall prove that (notice the uniformity in x)

$$\|A_T f - m(f)\|_\infty \xrightarrow{T \rightarrow \infty} 0 \quad \text{for every } f \in C(\mathbb{T}^2),$$

Lemma 0.3. *For every $T > 0$, every $h \in C(\mathbb{T}^2)$ and every $x \in \mathbb{T}^2$,*

$$|(A_T h)(x)| \leq \|h\|_\infty \quad \text{and} \quad |m(h)| \leq \|h\|_\infty.$$

In particular, A_T has operator norm 1 with respect to the sup-norm $\|\cdot\|_\infty$ on $C(\mathbb{T}^2)$.

Proof. Both are the triangle inequality for an average: since $\frac{1}{T} \int_0^T dt = 1$ and $m(\mathbb{T}^2) = 1$,

$$|(A_T h)(x)| \leq \frac{1}{T} \int_0^T |h(x + tv)| dt \leq \|h\|_\infty, \quad |m(h)| \leq \int_{\mathbb{T}^2} |h| dm \leq \|h\|_\infty. \quad \square$$

Both bounds are uniform in T and in x , which is what will allow us to choose an approximation to f once and for all and then let $T \rightarrow \infty$.

Step 1: characters. Let $n \in \mathbb{Z}^2$. If $n = 0$ then $e_0 \equiv 1$ and $A_T e_0 = 1 = m(e_0)$ for every T . If $n \neq 0$ then $\langle n, v \rangle \neq 0$ by Lemma 0.2, so by (0.1)

$$(A_T e_n)(x) = e_n(x) \cdot \frac{1}{T} \int_0^T e^{2\pi i t \langle n, v \rangle} dt = e_n(x) \cdot \frac{1}{T} \cdot \frac{e^{2\pi i T \langle n, v \rangle} - 1}{2\pi i \langle n, v \rangle},$$

whence, using $|e_n(x)| = 1$ and $|e^{i\theta} - 1| \leq 2$,

$$\|A_T e_n - m(e_n)\|_\infty = \|A_T e_n\|_\infty \leq \frac{1}{\pi T |\langle n, v \rangle|} \xrightarrow{T \rightarrow \infty} 0.$$

By linearity of A_T and of m , the same holds for every finite linear combination of characters:

$$(0.2) \quad \|A_T p - m(p)\|_\infty \xrightarrow{T \rightarrow \infty} 0 \quad \text{for every trigonometric polynomial } p.$$

Step 2: from polynomials to all of $C(\mathbb{T}^2)$. Let $f \in C(\mathbb{T}^2)$ and $\varepsilon > 0$. By Stone–Weierstrass there is a trigonometric polynomial p with $\|f - p\|_\infty < \varepsilon$. For every $T > 0$ and every $x \in \mathbb{T}^2$, the triangle inequality gives

$$|(A_T f)(x) - m(f)| \leq |(A_T(f - p))(x)| + |(A_T p)(x) - m(p)| + |m(p - f)|,$$

where we used $A_T f - A_T p = A_T(f - p)$ and $m(p) - m(f) = m(p - f)$, both by linearity. By Lemma 0.3 applied to $h = f - p$, the first and third terms are each $< \varepsilon$, uniformly in T and x . Taking the supremum over x ,

$$\|A_T f - m(f)\|_\infty \leq 2\varepsilon + \|A_T p - m(p)\|_\infty,$$

and the last term tends to 0 by (0.2). Hence

$$\limsup_{T \rightarrow \infty} \|A_T f - m(f)\|_\infty \leq 2\varepsilon,$$

and since $\varepsilon > 0$ was arbitrary the limit is 0. This is the assertion of (b), with convergence uniform in x .

Invariant measures. Let μ be a φ -invariant Borel probability measure. Applying invariance to $f = e_{-n}$ and using (0.1),

$$\widehat{\mu}(n) = \int e_{-n} \circ \varphi_t d\mu = e^{-2\pi i t \langle n, v \rangle} \widehat{\mu}(n) \quad \text{for all } t \in \mathbb{R},$$

so

$$(0.3) \quad \langle n, v \rangle \neq 0 \implies \widehat{\mu}(n) = 0.$$

In case (b), (0.3) kills every coefficient with $n \neq 0$ and $\widehat{\mu}(0) = 1$, so $\widehat{\mu} = \widehat{m}$ and $\mu = m$. In particular there is only one invariant probability measure at all, ergodic or not; this is the phenomenon called *unique ergodicity*, and it is what makes the convergence in (b) hold for *every* point rather than almost every point.

In case (a), write $v = \lambda w$ with w primitive and let $n_0 = (-w_2, w_1)$. Then $\langle n, v \rangle = 0$ exactly for $n \in \mathbb{Z}n_0$, so by (0.3) the coefficients $\widehat{\mu}(n)$ vanish off $\mathbb{Z}n_0$. Consequently, for any $y \in \mathbb{T}^2$ with $\langle n_0, y \rangle \in \mathbb{Z}$ the translate $\mu(\cdot - y)$ has the same Fourier coefficients as μ , hence equals μ : that is, μ is invariant under the circle subgroup Y . The quotient $\mathbb{T}^2/Y$ is a circle on which the flow acts trivially, so every Borel subset of $\mathbb{T}^2/Y$ pulls back to a φ -invariant subset of $\mathbb{T}^2$; if μ is ergodic, all these have measure 0 or 1, and the image of μ in $\mathbb{T}^2/Y$ is a point mass. Thus μ is supported on a single coset $x + Y$ and is Y -invariant there, i.e. it is the arclength measure of that periodic orbit. □

0.2. Three problems. The interactions of homogeneous dynamics with other fields are very rich, and in this section we will consider three deep number-theoretic questions that at first sight have nothing to do with dynamics, and which we will meet again in the course.

Problem A (Approximation and geodesics). *Dirichlet's theorem states that for every irrational $\alpha \in \mathbb{R}$ the inequality*

$$\left| \alpha - \frac{p}{q} \right| \leq \frac{1}{q^2}$$

has infinitely many solutions $(p, q) \in \mathbb{Z} \times \mathbb{N}$. Neither the exponent nor the constant can be improved for all α at once: Hurwitz showed that the constant 1 may be replaced by $1/\sqrt{5}$ and by nothing smaller, and the exponent 2 cannot be raised at all

as the following class of numbers shows. Call $\alpha \in \mathbb{R}$ badly approximable if there is $c > 0$ with

$$\left| \alpha - \frac{p}{q} \right| \geq \frac{c}{q^2} \quad \text{for all } p \in \mathbb{Z}, q \in \mathbb{N}.$$

These are the numbers that are as badly approximable by rationals as Dirichlet's theorem permits. Which numbers are they, how many are there, and is there a structural description?

Problem A has a classical answer in terms of continued fractions (α is badly approximable exactly when its partial quotients are bounded), but the finer questions — how large is the set of such α , and what distinguishes them structurally? — are not classical.

Problem B (Values of quadratic forms). *Let Q be a real quadratic form in n variables, non-degenerate and indefinite, and not a scalar multiple of a form with rational coefficients. Is the set of values*

$$Q(\mathbb{Z}^n) = \{Q(v) : v \in \mathbb{Z}^n\} \subseteq \mathbb{R}$$

dense in $\mathbb{R}$?

Problem B was raised by Oppenheim in 1929;¹ it is false for $n = 2$, was expected to be true for $n \geq 3$, and resisted the methods of analytic number theory for nearly sixty years, until Margulis proved it in 1986 by dynamical means.

Problem C (Littlewood, 1930). *Write $\|y\|$ for the distance from $y \in \mathbb{R}$ to the nearest integer. Is it true that*

$$\liminf_{q \rightarrow \infty} q \|q\alpha\| \|q\beta\| = 0 \quad \text{for all } \alpha, \beta \in \mathbb{R}?$$

Problem C is open today; the strongest evidence for it is a theorem of Einsiedler, Katok and Lindenstrauss (2006), also proved by dynamical means, which says that the set of pairs (α, β) for which the conjecture fails has Hausdorff dimension 0.

0.3. Their dynamical reformulation. Each of the three problems becomes, after a change of language, a question about a single orbit in a space of lattices. We describe the translation informally here; all the new notions will be introduced carefully later in the course.

The space is in all three problems the same. Let

$$X_d = \mathrm{SL}_d(\mathbb{R}) / \mathrm{SL}_d(\mathbb{Z}) \cong \{ \text{lattices } \Lambda \subseteq \mathbb{R}^d \text{ with } \mathrm{covol}(\Lambda) = 1 \},$$

the identification being $g \mathrm{SL}_d(\mathbb{Z}) \mapsto g\mathbb{Z}^d$, and let $\mathrm{SL}_d(\mathbb{R})$ act on X_d by acting linearly on lattices. Write

$$\lambda_1(\Lambda) = \min_{v \in \Lambda \setminus \{0\}} \|v\|$$

for the length of a shortest non-zero vector. The one fact we need is *Mahler's criterion*: a set of unimodular lattices is relatively compact in X_d if and only if λ_1 is bounded away from 0 on it. In other words,

leaving every compact subset of $X_d \leftrightarrow$ acquiring arbitrarily short vectors,

and this is the insight on which all three translations are based: a Diophantine inequality is a statement about short vectors in a lattice, and hence a statement about the position of a point of X_d .

¹Oppenheim's original conjecture concerned $n \geq 5$; the stronger form for $n \geq 3$, which is the one now called the Oppenheim conjecture, was formulated later by Davenport and Oppenheim.

Problem A: bounded geodesics. It will be convenient to write

$$\|y\| = \min_{p \in \mathbb{Z}} |y - p| \quad (y \in \mathbb{R}), \quad c(\alpha) = \inf_{q \geq 1} q \|q\alpha\|,$$

so that, dividing the defining inequality of Problem A by q , the number α is badly approximable precisely when $c(\alpha) > 0$.

Take $d = 2$, let

$$\Lambda_\alpha = \{(q, p - q\alpha) : p, q \in \mathbb{Z}\},$$

a unimodular lattice, and let the diagonal group $a_t = \text{diag}(e^{-t/2}, e^{t/2})$ act on it. For $v = (q, p - q\alpha)$,

$$\|a_t v\|^2 = e^{-t} q^2 + e^t (p - q\alpha)^2 \geq 2q |p - q\alpha|,$$

with equality for the t that balances the two terms. So $a_t \Lambda_\alpha$ has a short vector at some time t precisely when there are p, q with $q |q\alpha - p|$ small — that is, precisely when α admits a good rational approximation. Combined with Mahler's criterion:

$$\alpha \text{ is badly approximable} \iff \text{the forward orbit } \{a_t \Lambda_\alpha : t \geq 0\} \text{ is bounded in } X_2.$$

In more geometric terms, under the identification of X_2 with the unit tangent bundle of the modular surface, and of the flow (a_t) with the geodesic flow, this says that α is badly approximable exactly when the corresponding geodesic ray does not go arbitrarily far up the cusp.

Let us make this precise. Doing so costs one page, and it comes with a bonus: the same argument yields the classical description of the badly approximable numbers in terms of continued fractions, so that all three descriptions — Diophantine, dynamical, and by continued fractions — appear as one statement.

We assume from now on that α is irrational; for rational α the lattice Λ_α contains a non-zero vector $(q, 0)$, whose length under a_t tends to 0, so that the orbit is not merely unbounded but divergent. Let $\alpha = [b_0; b_1, b_2, \dots]$ be the continued fraction expansion, with convergents p_k/q_k .² We use the following standard facts, for which we refer to [EW1, Ch. 3] or to any textbook on continued fractions:

- (CF1) $q_{-1} = 0, q_0 = 1$ and $q_{k+1} = b_{k+1}q_k + q_{k-1}$ for $k \geq 0$; in particular $(q_k)_{k \geq 0}$ is non-decreasing and tends to infinity;
- (CF2) $p_k q_{k-1} - p_{k-1} q_k = (-1)^{k-1}$;
- (CF3) the numbers $q_k \alpha - p_k$ are non-zero, alternate in sign, and satisfy

$$\frac{1}{q_{k+1} + q_k} < |q_k \alpha - p_k| < \frac{1}{q_{k+1}}.$$

Beyond these we need one classical property of the convergents: among all denominators below q_{k+1} , none approximates better than q_k . Its proof is short, so we include it.

Lemma 0.4 (Best approximation). *Let $k \geq 0$, let $q \in \mathbb{N}$ with $q < q_{k+1}$ and let $p \in \mathbb{Z}$. Then*

$$|q\alpha - p| \geq |q_k \alpha - p_k|.$$

Proof. By (CF2) the matrix $\begin{pmatrix} p_k & p_{k+1} \\ q_k & q_{k+1} \end{pmatrix}$ has determinant ± 1 , so there are integers x, y with

$$p = xp_k + yp_{k+1}, \quad q = xq_k + yq_{k+1}.$$

If $y = 0$ then $q = xq_k$ with $x \neq 0$, so $|q\alpha - p| = |x| |q_k \alpha - p_k| \geq |q_k \alpha - p_k|$. If $x = 0$ then $q = yq_{k+1} \geq q_{k+1}$, contrary to assumption. So assume $x, y \neq 0$. From

²The partial quotients are usually denoted a_k ; we write b_k to avoid a collision with the diagonal flow a_t .

$1 \leq q < q_{k+1}$ and $q = xq_k + yq_{k+1}$ we see that x and y must have opposite signs. By (CF3) the numbers $q_k\alpha - p_k$ and $q_{k+1}\alpha - p_{k+1}$ also have opposite signs. Hence the two terms in

$$q\alpha - p = x(q_k\alpha - p_k) + y(q_{k+1}\alpha - p_{k+1})$$

have the *same* sign, so no cancellation occurs and

$$|q\alpha - p| = |x| |q_k\alpha - p_k| + |y| |q_{k+1}\alpha - p_{k+1}| \geq |q_k\alpha - p_k|. \quad \square$$

Proposition 0.5. *Let $\alpha \in \mathbb{R}$ be irrational. The following are equivalent.*

- (i) α is badly approximable, i.e. $c(\alpha) > 0$;
- (ii) the forward orbit $\{a_t\Lambda_\alpha : t \geq 0\}$ is bounded in X_2 ;
- (iii) the partial quotients $b_1, b_2, \dots$ of α are bounded.

Quantitatively, if $M = \sup_{k \geq 1} b_k$ then

$$(0.4) \quad \frac{1}{M+2} \leq c(\alpha) \leq \frac{1}{M},$$

and if $c = c(\alpha) > 0$ then $\lambda_1(a_t\Lambda_\alpha) \geq \min(1, \sqrt{2c})$ for all $t \geq 0$.

Proof. (i) $\Rightarrow$ (ii). Let $c = c(\alpha) > 0$, let $t \geq 0$ and let $v = (q, p - q\alpha)$ be a non-zero element of Λ_α . If $q = 0$ then $p \neq 0$ and $\|a_tv\| = e^{t/2}|p| \geq 1$. If $q \neq 0$ we may assume $q \geq 1$, after replacing (p, q) by $(-p, -q)$, which changes neither $\|a_tv\|$ nor the hypothesis; then by the displayed inequality above and by $|p - q\alpha| \geq \|q\alpha\|$,

$$\|a_tv\|^2 \geq 2q|p - q\alpha| \geq 2q\|q\alpha\| \geq 2c.$$

Hence $\lambda_1(a_t\Lambda_\alpha) \geq \min(1, \sqrt{2c})$ for every $t \geq 0$, and the forward orbit is bounded by Mahler's criterion.

(ii) $\Rightarrow$ (i). We prove the contrapositive. Suppose $c(\alpha) = 0$ and choose $q_n \geq 1$ and $q_n \in \mathbb{Z}$ with $|q_n - q_n\alpha| = \|q_n\alpha\|$ and $\delta_n := q_n\|q_n\alpha\| \rightarrow 0$. Since α is irrational we have $\|q_n\alpha\| > 0$, and $\|q_n\alpha\| \leq \delta_n \rightarrow 0$ while $q_n \geq 1$, so the numbers

$$t_n := \log \frac{q_n}{\|q_n\alpha\|}$$

are positive for all large n . These are exactly the balancing times mentioned above: writing $v_n = (q_n, q_n - q_n\alpha)$,

$$e^{-t_n}q_n^2 = q_n\|q_n\alpha\| = \delta_n, \quad e^{t_n}\|q_n\alpha\|^2 = q_n\|q_n\alpha\| = \delta_n,$$

so $\|a_{t_n}v_n\| = \sqrt{2\delta_n} \rightarrow 0$. Thus $\lambda_1(a_{t_n}\Lambda_\alpha) \rightarrow 0$ and, by Mahler's criterion again, the forward orbit leaves every compact set.

(iii) $\Rightarrow$ (i). Suppose $b_k \leq M$ for all $k \geq 1$, and let $q \geq 1$ and $p \in \mathbb{Z}$. Choose $k \geq 0$ maximal with $q_k \leq q$; this is possible by (CF1), and then $q < q_{k+1}$. By Lemma 0.4 and (CF3),

$$q|q\alpha - p| \geq q_k|q_k\alpha - p_k| > \frac{q_k}{q_{k+1} + q_k}.$$

Now $q_{k+1} + q_k = b_{k+1}q_k + q_{k-1} + q_k \leq (M+2)q_k$, using $q_{k-1} \leq q_k$, so the right-hand side is at least $1/(M+2)$. As p and q were arbitrary, $c(\alpha) \geq 1/(M+2) > 0$.

(i) $\Rightarrow$ (iii). By (CF3) and (CF1), for every $k \geq 0$,

$$c(\alpha) \leq q_k\|q_k\alpha\| \leq q_k|q_k\alpha - p_k| < \frac{q_k}{q_{k+1}} \leq \frac{q_k}{b_{k+1}q_k} = \frac{1}{b_{k+1}}.$$

So $b_{k+1} \leq 1/c(\alpha)$ for every k : if $c(\alpha) > 0$ the partial quotients are bounded by $1/c(\alpha)$. This also proves the second inequality in (0.4), the first having been proved in the previous paragraph. $\square$

Remark 0.6. *The two-sided estimate (0.4) is worth remembering: the size of the partial quotients and the quality of the best approximations are the same quantity up to a bounded factor, and by Proposition 0.5 both measure how far up the cusp the corresponding geodesic ray travels. A large partial quotient b_{k+1} is a single very good rational approximation p_k/q_k , which is a single deep excursion into the cusp; and the excursion is deep in proportion to b_{k+1} .*

Example 0.7. *The golden ratio $\varphi = \frac{1+\sqrt{5}}{2} = [1; 1, 1, \dots]$ has all partial quotients equal to 1, so $M = 1$ and (0.4) gives $\frac{1}{3} \leq c(\varphi) \leq 1$. The exact value is $c(\varphi) = 1/\sqrt{5}$, and by Hurwitz's theorem no irrational number has a larger one: in this sense φ is the worst approximable of all irrationals, and its geodesic is the one that stays lowest in the cusp.*

We can now return to the three questions raised in Problem A, all of which have become questions about geodesics on the modular surface. *Which numbers are badly approximable?* Proposition 0.5 answers this in terms of continued fractions; dynamically, the quadratic irrationals — those with an eventually periodic expansion, by a theorem of Euler and Lagrange — are exactly the α whose geodesic is periodic, as we shall see in Section 2.5. *How many are there?* The set of bounded geodesics has measure zero but full Hausdorff dimension, so the answer depends entirely on how one counts. *Is there a structural description?* What is being asked for is a classification of the orbit closures of the geodesic flow — and, as we shall see, no such classification exists in any reasonable sense.

Problem B: a dense orbit. Fix a standard form Q_0 of the signature of Q . Every non-degenerate Q of that signature can be written as $Q = \lambda \cdot Q_0 \circ g$ with $\lambda \neq 0$ and $g \in \mathrm{SL}_n(\mathbb{R})$, so that

$$Q(\mathbb{Z}^n) = \lambda Q_0(g\mathbb{Z}^n) = \lambda Q_0(\Lambda), \quad \Lambda := g\mathbb{Z}^n \in X_n.$$

The question about the form Q has become a question about the fixed form Q_0 evaluated on a variable lattice. Now let

$$H = \mathrm{SO}(Q_0)^\circ \leq \mathrm{SL}_n(\mathbb{R})$$

be the group preserving Q_0 . For $h \in H$ we have $Q_0(h\Lambda) = Q_0(\Lambda)$: the set of values is constant along the H -orbit of the point $\Lambda \in X_n$. This is what makes the problem dynamical, and it gives at once one half of the translation. Suppose the orbit ΛH is dense in X_n , and let $c \in \mathbb{R}$. Since Q_0 is indefinite, there is a vector w with $Q_0(w) = c$; choose any unimodular lattice Λ' containing w . Every lattice sufficiently close to Λ' contains a vector close to w , hence takes a value close to c ; and by density some $h\Lambda$ is that close. As $Q_0(h\Lambda) = Q_0(\Lambda)$, the set $Q_0(\Lambda)$ meets every neighbourhood of c . Thus

$$\Lambda H \text{ dense in } X_n \implies Q(\mathbb{Z}^n) \text{ dense in } \mathbb{R}.$$

The other half is the arithmetic one: if Q is proportional to a rational form, then $H \cap \mathrm{SL}_n(\mathbb{Z})$ is large — a lattice in H — and the orbit ΛH is closed rather than dense. So Oppenheim's conjecture is exactly the assertion that these are the only two possibilities, which is a statement of the form

$$\text{every } H\text{-orbit is either closed or dense.}$$

This is what Margulis proved for $n = 3$, and it is a special case of Ratner's theorem. Two remarks are in order. First, $H = \mathrm{SO}(Q_0)^\circ$ is generated by unipotent one-parameter subgroups as soon as $n \geq 3$, which is why the rigidity theory applies. Second, for $n = 2$ the group $\mathrm{SO}(1, 1)^\circ$ is conjugate to the diagonal group A of the

previous paragraph, and there the assertion is false — indeed $Q(m, n) = (m - \alpha n)n$ with α badly approximable has $|Q|$ bounded away from 0 on $\mathbb{Z}^2 \setminus \{0\}$, by the computation above. So the counterexamples to Oppenheim in two variables *are* the badly approximable numbers of Problem A: the first two problems are the same problem, seen for $n = 2$ and $n \geq 3$.

Problem C: a bounded orbit in rank two. Take $d = 3$ and, given α, β , the unimodular lattice

$$\Lambda_{\alpha, \beta} = \begin{pmatrix} 1 & & \\ -\alpha & 1 & \\ -\beta & & 1 \end{pmatrix} \mathbb{Z}^3 = \{(q, p_1 - q\alpha, p_2 - q\beta) : q, p_1, p_2 \in \mathbb{Z}\},$$

acted on by the full diagonal group

$$A_3 = \{\text{diag}(e^{t_1}, e^{t_2}, e^{t_3}) : t_1 + t_2 + t_3 = 0\}.$$

The product of the coordinates $|v_1 v_2 v_3|$ is invariant under A_3 , and on $\Lambda_{\alpha, \beta}$ it takes the values $q |p_1 - q\alpha| |p_2 - q\beta|$, whose infimum over p_1, p_2 is Littlewood's quantity $q \|q\alpha\| \|q\beta\|$. On the other hand, the same balancing argument as for $d = 2$ — now with three terms — gives

$$\inf_{a \in A_3} \lambda_1(a\Lambda)^2 = 3 \left(\inf_{v \in \Lambda \setminus \{0\}} |v_1 v_2 v_3| \right)^{2/3},$$

so that by Mahler's criterion

Littlewood's conjecture fails for $(\alpha, \beta) \iff$ the A_3 -orbit of $\Lambda_{\alpha, \beta}$ is bounded in X_3 .

A conjecture quantified over all integers q has become a statement about the orbits of one abelian group acting on one space. The difference from Problem A is that A_3 has rank 2 rather than rank 1: it is a two-parameter group of commuting flows, and this is precisely the situation in which rigidity is expected to be restored. The theorem of Einsiedler, Katok and Lindenstrauss quoted above says that the set of $x \in X_3$ with bounded A_3 -orbit has Hausdorff dimension 0; Littlewood's conjecture asks that the only such orbits be the compact ones.

Summary.

Problem	space	group	rank	the dynamical statement
A	X_2	A	1	which orbits are bounded?
B	X_n	$\text{SO}(Q_0)^\circ$	—	is every orbit closed or dense?
C	X_3	A_3	2	which orbits are bounded?

The pattern to retain is the dictionary

$$\begin{aligned} \text{bounded orbit} &\longleftrightarrow \text{Diophantine exception,} \\ \text{closed orbit} &\longleftrightarrow \text{rational object,} \\ \text{dense orbit} &\longleftrightarrow \text{density of values,} \end{aligned}$$

together with the observation that the answers depend dramatically on which group acts. For a group generated by unipotents the behaviour is rigid and everything is known; for a rank-one diagonal group it is chaotic and no classification exists; for a higher-rank diagonal group rigidity is expected but unproved.

0.4. Ratner's theorems. In §0.1 we used the two-torus as a toy model and answered, in that setting, typical questions of homogeneous dynamics: orbit closures, invariant measures, equidistribution. That example is not merely an analogy. Since $\mathbb{R}^2$ is a unipotent group and $\mathbb{Z}^2$ a lattice in it, Theorem 0.1 is one of the simplest cases of Ratner's theorems, which we state in this section. Three features of the answer are worth naming, because all three will reappear.

- (i) *Dichotomy.* There is no intermediate behaviour: an orbit is either closed or dense. Nothing fractal, nothing partially dense.
- (ii) *Algebraicity.* Both the orbit closures and the ergodic invariant measures are described by a closed subgroup: they are cosets of Y , respectively the Haar measures of such cosets. The list of possibilities is a list of subgroups, not a list of dynamical behaviours.
- (iii) *Arithmetic.* Which of the two cases occurs is decided by a rationality condition on v .

What does *not* generalise is the proof. It used that $X = \mathbb{T}^2$ is itself a compact abelian group, so that the flow is diagonalised by characters. When G/Γ is a quotient of a non-abelian group it is not a group at all, there are no characters, and X need not be compact — so mass can escape to infinity, a possibility that does not even arise on $\mathbb{T}^2$. Ratner's theorems assert the same conclusion in that generality, and their proofs are correspondingly harder.

We first state them in the case that will occupy us later in this course, namely $G = \mathrm{SL}_2(\mathbb{R})$, $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, $X_2 = \mathrm{SL}_2(\mathbb{R})/\Gamma$ with its invariant probability measure m_{X_2} , and the horocycle subgroup $U = U^-$ of §2.6,

$$U = \left\{ u_s = \begin{pmatrix} 1 & s \\ & 1 \end{pmatrix} : s \in \mathbb{R} \right\}, \quad u_s \cdot x = xu_s^{-1};$$

as a set the orbit $U \cdot x$ is the same as xU . The reader should compare each item with Theorem 0.1.

Theorem 0.8 (Ratner for X_2 ; Hedlund, Furstenberg, Dani, Dani–Smillie). *Let $x \in X_2$.*

- (i) (Orbit closures.) *Either the orbit $U \cdot x$ is periodic, and then it is closed, or it is dense in X_2 .*
- (ii) (Invariant measures.) *Every U -invariant ergodic Borel probability measure on X_2 is either m_{X_2} or the normalised arclength measure on a periodic U -orbit.*
- (iii) (Equidistribution.) *If $U \cdot x$ is not periodic, then for every $f \in C_c(X_2)$*

$$\frac{1}{S} \int_0^S f(u_s \cdot x) ds \xrightarrow{S \rightarrow \infty} \int_{X_2} f dm_{X_2}.$$

The exceptional orbits are known explicitly, and it is worth recording what they look like, since we shall meet them again in §3. By §2.7 the orbit $Ug\Gamma$ is periodic if and only if $g \cdot \infty \in \mathbb{Q} \cup \{\infty\}$, and in that case $Ug\Gamma = Ua_t\Gamma$ for some $t \in \mathbb{R}$. Writing

$$P_t := Ua_t^{-1}\Gamma \quad (t \in \mathbb{R}),$$

the relation $a_t u_s a_t^{-1} = u_{e^{-t}s}$ gives $u_s a_t^{-1}\Gamma = a_t^{-1}\Gamma$ if and only if $e^{-t}s \in \mathbb{Z}$, so P_t is periodic of period e^t . Since the period is an invariant of the orbit, the P_t are pairwise distinct. In particular there are uncountably many of them.

Two differences from the torus deserve to be pointed out at once. The first is the one just described: on $\mathbb{T}^2$ the periodic orbits, when they occur, occur for *every*



FIGURE 2. Two members of the family $\{P_t\}$, projected to the modular surface and drawn in the fundamental domain E (the two vertical sides are identified, and so are the two halves of the circular arc). Left: the closed horocycle at height $y = 1.95$, of hyperbolic length $1/1.95$. Right: the closed horocycle at height $y = 0.01$, of length 100, cut into arcs by the folding; each arc is a piece of a Euclidean circle tangent to the real axis. It is on its way to equidistributing with respect to the hyperbolic area, which is why the arcs crowd near the bottom. About a third of the curve lies above the dashed line, on its way into the cusp.

point, whereas here they form a thin but uncountable family sitting inside a space where the typical orbit is dense. Nevertheless the exceptional orbits are not lost to the theory: as their period tends to infinity they equidistribute towards m_{X_2} , which is the theorem we prove in §3. The second difference is that X_2 is not compact, so a priori the averages in (iii) could lose mass into the cusp and converge to a sub-probability measure. Ruling this out — *non-divergence* — is a genuine part of the proof and has no analogue in Theorem 0.1.

It is equally important to know what the theorem does not say. Replace U by the diagonal group A of §2.3 and all three statements become false: there are orbits that are neither periodic nor dense, there are uncountably many ergodic invariant measures, and orbit closures can be fractal. The difference is that u_s is unipotent while a_t is diagonalisable, and this is precisely the hypothesis in the general theorem.

Theorem 0.9 (Ratner, 1991). *Let G be a connected Lie group, $\Gamma \leq G$ a lattice, $X = G/\Gamma$, and let $H \leq G$ be a closed connected subgroup generated by one-parameter unipotent subgroups. Then:*

- (i) *for every $x \in X$ there is a closed subgroup L with $H \leq L \leq G$ such that*

$$\overline{H \cdot x} = L \cdot x,$$

and $L \cdot x$ is closed and carries a finite L -invariant measure $m_{L \cdot x}$;

- (ii) *every H -invariant ergodic Borel probability measure on X is of the form $m_{L \cdot x}$ for such a pair (L, x) ;*

- (iii) *if $H = \{u_t\}$ is a one-parameter unipotent subgroup, then every orbit equidistributes in its closure: with L as in (i),*

$$\frac{1}{T} \int_0^T f(u_t \cdot x) dt \xrightarrow{T \rightarrow \infty} \int f dm_{L \cdot x} \quad (f \in C_c(X)).$$

Measures and orbit closures are thus *homogeneous*: each is attached to a subgroup. This is exactly the conclusion of Theorem 0.1, with “coset of a circle subgroup” replaced by “closed orbit of a closed subgroup”, and it is the reason such

results are called rigidity theorems — the invariant objects are far too few to be described dynamically, and must be described algebraically.

0.5. Oppenheim’s conjecture as a consequence. We can now see Problem B fall out of Theorem 0.9. Recall the reformulation of §0.2: writing $Q = \lambda \cdot Q_0 \circ g$ with Q_0 a fixed form of the same signature, $\lambda \neq 0$ and $g \in \mathrm{SL}_n(\mathbb{R})$, and setting $\Lambda = g\mathbb{Z}^n \in X_n$, we have

$$Q(\mathbb{Z}^n) = \lambda Q_0(\Lambda),$$

and the group $H = \mathrm{SO}(Q_0)^\circ$ leaves $Q_0(\Lambda)$ unchanged along its orbit. We showed there that if $H \cdot \Lambda$ is dense in X_n then $Q(\mathbb{Z}^n)$ is dense in $\mathbb{R}$. So everything comes down to identifying the orbit closure, and for that we quote two classical facts about the groups involved.

Fact 0.10. *For $n \geq 3$ and Q_0 non-degenerate indefinite, the group $H = \mathrm{SO}(Q_0)^\circ$ is semisimple with no compact factors, hence is generated by one-parameter unipotent subgroups. Moreover H is a maximal connected proper subgroup of $\mathrm{SL}_n(\mathbb{R})$.*

Fact 0.11 (Borel density theorem). *If H is as above and $\Delta \leq H$ is a lattice, then Δ is Zariski-dense in H .*

Corollary 0.12 (Oppenheim’s conjecture). *Let $n \geq 3$ and let Q be a non-degenerate indefinite real quadratic form in n variables which is not a scalar multiple of a form with rational coefficients. Then $Q(\mathbb{Z}^n)$ is dense in $\mathbb{R}$.*

Proof. By Fact 0.10 the group H is generated by unipotent one-parameter subgroups, so Theorem 0.9 applies: $\overline{H \cdot \Lambda} = L \cdot \Lambda$ for a closed subgroup $H \leq L \leq \mathrm{SL}_n(\mathbb{R})$ with $L \cdot \Lambda$ of finite volume. Replacing L by its identity component changes nothing in what follows, so by the maximality in Fact 0.10 there are only two possibilities.

If $L = \mathrm{SL}_n(\mathbb{R})$, the orbit $H \cdot \Lambda$ is dense in X_n and we are done by the argument of §0.2.

Suppose then $L = H$, so that the orbit $H \cdot \Lambda$ is itself closed and carries a finite H -invariant measure. Equivalently, the stabiliser

$$\Delta = \mathrm{Stab}_H(\Lambda) = \{h \in H : h\Lambda = \Lambda\}$$

is a lattice in H . Every $h \in \Delta$ preserves the lattice $\Lambda = g\mathbb{Z}^n$, so $g^{-1}\Delta g \subseteq \mathrm{SL}_n(\mathbb{Z})$. By Fact 0.11, Δ is Zariski-dense in H , so $g^{-1}Hg$ is the Zariski closure of a subgroup of $\mathrm{SL}_n(\mathbb{Z})$ and is therefore an algebraic group defined over $\mathbb{Q}$. But

$$g^{-1}Hg = \mathrm{SO}(Q_0 \circ g)^\circ = \mathrm{SO}(Q)^\circ,$$

and the quadratic form preserved by a $\mathbb{Q}$ -defined orthogonal group is, up to a scalar, defined over $\mathbb{Q}$. Hence Q is proportional to a rational form, contrary to hypothesis. So this case does not occur. $\square$

0.6. Equidistribution of long closed horocycles. Theorem 0.8 leaves one case untreated. Part (iii) says nothing about the periodic orbits, and for a good reason: a periodic orbit carries its own invariant measure, the arclength measure, and a point moving along it distributes according to that measure and not according to m_{X_2} . So the exceptional orbits of part (i) really are exceptional. On the other hand, they come in a one-parameter family with periods tending to infinity, and one may ask what happens to them in that limit.

The family has a simple geometric description. Since $a_t u_s a_t^{-1} = u_{e^{-t}s}$ we have $U a_t \Gamma = a_t U \Gamma$ for all $t \in \mathbb{R}$. Note that $P_0 = U \Gamma$ has period 1 and that

$$P_t = a_t P_0,$$

has period e^t . In fact, *all* the closed horocycles are translates by the geodesic flow of P_0 , the horocycle through $(i, \uparrow)$. In the upper half-plane picture, P_0 is the segment at height 1 and P_t is the segment at height e^{-t} : the geodesic flow pushes the horocycle down towards the real axis, and as it descends it becomes longer — of length e^t — and wraps more and more times around the modular surface. Figure 3 shows P_0 and one of its translates folded back into the fundamental domain.



FIGURE 3. Left: the standard closed horocycle P_0 , the segment at height 1 through i , of length 1. Right: its translate P_t with $e^{-t} = 0.1$, of length 10, pulled down by the geodesic flow and folded back into E ; the folding cuts it into arcs, each a piece of a Euclidean circle tangent to the real axis. Compare Figure 7, which is the same picture for length 100.

The answer is that in the limit the exceptional orbits cease to be exceptional.

Theorem 0.13 (Equidistribution of long closed horocycles). *For every $f \in C_c(X_2)$,*

$$e^{-t} \int_0^{e^t} f(u_s a_t \Gamma) ds \xrightarrow{t \rightarrow \infty} \int_{X_2} f dm_{X_2}.$$

We prove this later in the course. It completes the description begun in Theorem 0.8: every orbit either equidistributes, or is periodic and belongs to a family which equidistributes in the limit.

The proof is a beautiful argument of Margulis, known as the *banana trick*. The obstacle is that μ_t lives on a one-dimensional subset of the three-dimensional space X_2 , so it is singular with respect to m_{X_2} and no amount of mixing applies to it directly. The remedy is to thicken P_t in the two transverse directions, by an amount chosen so that the thickened sets S_t satisfy

$$S_t = a_{-t} S_0 a_t;$$

this is possible because the transverse directions are contracted by the geodesic flow at exactly the rate at which the horocycle is stretched. The thickened measure is absolutely continuous, the average against it becomes a matrix coefficient of the geodesic flow, and mixing finishes the argument. The reader should note what this buys and what it does not: the argument is soft, and produces no rate of convergence whatsoever.

Rates require harder input, namely the spectral theory of the Laplacian on the modular surface. It is convenient to state the result downstairs, on $\mathbb{H}/\Gamma$ rather than on X_2 : for f smooth and compactly supported on the modular surface, and $y > 0$, put

$$I_y(f) = \int_0^1 f(x + iy) dx,$$

the average of f over the closed horocycle at height y , which is the projection of P_t for $e^{-t} = y$. Theorem 0.13 says that $I_y(f) \rightarrow \frac{3}{\pi} \int f dA$ as $y \rightarrow 0$, where $dA = y^{-2} dx dy$ and $\frac{\pi}{3} = \text{vol}(\mathbb{H}/\Gamma)$.

Theorem 0.14 (Sarnak [S]; Zagier [Za]). *Let f be smooth and compactly supported on $\mathbb{H}/\Gamma$.*

(i) *As $y \rightarrow 0$,*

$$I_y(f) = \frac{3}{\pi} \int_{\mathbb{H}/\Gamma} f dA + O_f(y^{1/2}).$$

(ii) *The Riemann hypothesis holds if and only if the exponent $1/2$ can be improved to $3/4$: that is, if and only if for every such f and every $\varepsilon > 0$,*

$$I_y(f) = \frac{3}{\pi} \int_{\mathbb{H}/\Gamma} f dA + O_{f,\varepsilon}(y^{3/4-\varepsilon}) \quad (y \rightarrow 0).$$

The mechanism behind (ii) is worth describing, even though we shall not prove it. Expand f spectrally, into its mean value, the Maass cusp forms, and the Eisenstein series $E(z, s)$ on the line $\text{Re } s = \frac{1}{2}$. A Maass cusp form has vanishing constant Fourier coefficient, so the cuspidal part of f contributes *nothing* to the horocycle average $I_y(f)$: the main term comes from the mean value, and the entire error term comes from the Eisenstein series. Now the constant Fourier coefficient of $E(z, s)$ is

$$y^s + \varphi(s) y^{1-s}, \quad \varphi(s) = \frac{\xi(2s-1)}{\xi(2s)},$$

where ξ is the completed Riemann zeta function. On the line $\text{Re } s = \frac{1}{2}$ both terms are of size $y^{1/2}$, which is where the exponent in (i) comes from. To do better one shifts the contour to the left; in doing so one picks up the poles of φ , which are the points $s = \rho/2$ with ρ a non-trivial zero of ζ , and each contributes a term of size

$$|y^{1-\rho/2}| = y^{1-\frac{1}{2}\text{Re } \rho}.$$

A zero with $\text{Re } \rho = \frac{1}{2}$ contributes $y^{3/4}$; a hypothetical zero with $\text{Re } \rho > \frac{1}{2}$ would contribute a strictly larger error. So the bound of (ii) holds exactly when no zero lies to the right of the critical line.

0.7. Classification of stationary measures. There is a second way of acting on a homogeneous space, and it will be the subject of the last part of the course, so we introduce it here. Instead of letting a subgroup act by a flow, fix a probability measure μ on the acting group and let the group act *at random*: from a point x , choose g with law μ and move to $g \cdot x$. Iterating with independent choices $g_1, g_2, \dots$ produces the trajectory

$$\omega_n = g_n \cdots g_1 \cdot x.$$

The object that replaces an invariant measure is the following.

Definition 0.15 (Stationary measure). Let a group G act on a space X and let μ be a Borel probability measure on G . A Borel probability measure ν on X is μ -stationary if it is invariant on average,

$$\mu * \nu := \int_G g_* \nu d\mu(g) = \nu.$$

It is *ergodic* if it is an extreme point of the convex set of μ -stationary probability measures.

Every G -invariant measure is μ -stationary, for any μ . The interest of the notion lies entirely in the fact that the converse can fail, and we shall see in Remark 0.17 that it does fail in the first example that matters. On the torus, however, it does not fail, and the proof is again a few lines of Fourier analysis.

Consider again the toy model $X = \mathbb{T}^2 = \mathbb{Z}^2 \backslash \mathbb{R}^2$ and let μ be a Borel probability measure on $G = \mathbb{R}^2$; the associated random walk moves from x to $x + v$ with v distributed according to μ . Writing $\pi: \mathbb{R}^2 \rightarrow \mathbb{T}^2$ for the projection, let

$$H := \overline{\langle \pi(\text{supp } \mu) \rangle} \subseteq \mathbb{T}^2$$

be the closed subgroup of $\mathbb{T}^2$ generated by the support of μ ,³ this is the group the walk can actually reach, and it plays the role played by the line $\mathbb{R}v$ in §0.1.

Theorem 0.16 (Stationary measures on $\mathbb{T}^2$). *Every ergodic μ -stationary probability measure on $\mathbb{T}^2$ is the Haar measure of a single coset of H . In particular every μ -stationary measure is invariant under translation by every element of $\text{supp } \mu$; and if $H = \mathbb{T}^2$, then Lebesgue measure is the unique μ -stationary probability measure.*

Proof sketch. As in §0.1 we use the characters $e_n(x) = e^{2\pi i \langle n, x \rangle}$, $n \in \mathbb{Z}^2$, and the Fourier coefficients $\widehat{\nu}(n) = \int e_{-n} d\nu$. For $v \in \mathbb{R}^2$ the push-forward of ν under $x \mapsto x + v$ has Fourier coefficients $e_{-n}(v) \widehat{\nu}(n)$, so integrating in v turns the definition of stationarity into a product:

$$(0.5) \quad \widehat{\mu * \nu}(n) = \widehat{\mu}(n) \widehat{\nu}(n), \quad \widehat{\mu}(n) = \int_{\mathbb{R}^2} e^{-2\pi i \langle n, v \rangle} d\mu(v).$$

Hence ν is stationary if and only if $\widehat{\nu}(n)(1 - \widehat{\mu}(n)) = 0$ for every n .

Now $|\widehat{\mu}(n)| \leq 1$, and $\widehat{\mu}(n) = 1$ forces the integrand to be constant: since $\text{Re}(1 - e^{-2\pi i \langle n, v \rangle}) \geq 0$ and integrates to 0, we get $\langle n, v \rangle \in \mathbb{Z}$ for μ -almost every v , hence for every $v \in \text{supp } \mu$ by continuity; since $n \in \mathbb{Z}^2$ the same holds on $\text{supp } \mu + \mathbb{Z}^2$, hence on the subgroup generated, and by continuity on its closure — that is, e_n is trivial on H . Conversely such an n clearly has $\widehat{\mu}(n) = 1$. Writing $H^\perp = \{n \in \mathbb{Z}^2 : e_n \equiv 1 \text{ on } H\}$ we have shown

$$(0.6) \quad \widehat{\nu}(n) = 0 \quad \text{for every } n \notin H^\perp.$$

From here the argument is the one used for case (a) of Theorem 0.1. If $y \in H$ then $e_{-n}(y) = 1$ for every $n \in H^\perp$, so by (0.6) the translate $\nu(\cdot - y)$ has the same Fourier coefficients as ν and therefore equals ν : the measure ν is H -invariant. In particular it is invariant under every element of $\text{supp } \mu$, which is the second assertion. The quotient $\mathbb{T}^2/H$ is a compact abelian group on which the walk acts trivially, because every step lies in H ; so if $A \subseteq \mathbb{T}^2$ is a union of H -cosets then $v + A = A$ for all $v \in \text{supp } \mu$, and the restrictions $\nu|_A$ and $\nu|_{\mathbb{T}^2 \setminus A}$ are again stationary. Ergodicity

³Both operations matter, and in this order: a step v and a step $v + \mathbb{Z}^2$ act identically on $\mathbb{T}^2$, and the closure must be taken downstairs, since for $\mu = \delta_{(\alpha, 0)}$ with α irrational the group generated in $\mathbb{R}^2$ is discrete while H is a circle.

therefore forces $\nu(A) \in \{0, 1\}$ for all such A , the image of ν in $\mathbb{T}^2/H$ is a point mass, and ν is the Haar measure of a single coset of H . If $H = \mathbb{T}^2$ there is only one coset, and only one stationary measure at all. $\square$



FIGURE 4. Take $\mu = \frac{1}{2}(\delta_v + \delta_w)$ with $v = (\alpha, 0)$ and $w = (0, \beta)$, $\alpha = \sqrt{2} - 1$, $\beta = \sqrt{3} - 1$. Left: the orbits through x of the two flows $\mathbb{R}v$ and $\mathbb{R}w$ separately — each is a closed circle, and neither is dense. Right: the first 600 points of one trajectory of the random walk driven by μ , which fills the torus. The walk sees the group generated by the support of μ , and that group is larger than either of the two lines.

The figure makes the point that the random walk is not a weaker object than a flow but a different one. For $v = (\alpha, 0)$ and $w = (0, \beta)$ with α, β irrational, both one-parameter flows $\mathbb{R}v$ and $\mathbb{R}w$ fall into case (a) of Theorem 0.1: every orbit is a closed circle. Yet $H = \mathbb{T}^2$ here, so by Theorem 0.16 the walk driven by $\mu = \frac{1}{2}(\delta_v + \delta_w)$ admits Lebesgue measure as its only stationary measure. Randomising has produced rigidity where each individual direction had none, because a walk explores the whole group generated by the support of μ rather than a single one-parameter subgroup.

Remark 0.17 (The example that makes the theory necessary). *Let $G = \mathrm{SL}_2(\mathbb{R})$ act on the projective line $\mathbb{P}(\mathbb{R}^2)$ and take*

$$\mu = \frac{1}{2}(\delta_{a_0} + \delta_{a_1}), \quad a_0 = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \quad a_1 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

Then $\mathbb{P}(\mathbb{R}^2)$ carries exactly one μ -stationary probability measure ν , it has no atoms, and it is not invariant under G — indeed $\mathbb{P}(\mathbb{R}^2)$ carries no G -invariant probability measure at all, since the action is transitive with non-compact stabilisers. This measure is the Furstenberg measure of the walk; it is typically singular with respect to Lebesgue measure, and it is the basic object of the theory of random walks on Lie groups. Nothing like it exists on $\mathbb{T}^2$, and its existence is why stationary measures need a theory of their own rather than a reduction to invariant ones.

The theorem of Benoist and Quint. The corresponding statement for X_2 is a theorem of Benoist and Quint, and it is one of the three results this course proves. Let μ be a compactly supported Borel probability measure on $G = \mathrm{SL}_2(\mathbb{R})$ and let

$$\Gamma_\mu = \langle \mathrm{supp} \mu \rangle \leq G$$

be the subgroup generated by its support. We assume that Γ_μ is *Zariski dense* in G : every polynomial function on G vanishing on Γ_μ vanishes identically. This is much weaker than density in the usual topology — $\mathrm{SL}_2(\mathbb{Z})$ is already Zariski dense in $\mathrm{SL}_2(\mathbb{R})$ — and it is the hypothesis that will play the role played by unipotency in §0.4.

Theorem 0.18 (Benoist–Quint [BQ1], 2011). *With μ as above, every ergodic μ -stationary Borel probability measure on X_2 is either the uniform measure on a finite Γ_μ -orbit, or the Haar measure m_{X_2} .*

The finite orbits are a genuine but harmless exception: there are only countably many of them, and they are the analogue of the periodic orbits in Theorem 0.8.

Remark 0.19 (Rigidity restored by randomising). *The force of Theorem 0.18 is seen by choosing μ badly. Nothing prevents $\text{supp } \mu$ from consisting of diagonalisable elements: take $\mu = \frac{1}{2}(\delta_{g_1} + \delta_{g_2})$ with g_1, g_2 hyperbolic and generating a Zariski dense subgroup. Each g_i on its own generates a cyclic subgroup of a conjugate of the diagonal group A , for which there is no classification of orbit closures at all. Yet the walk which alternates between the two at random admits no stationary measure beyond m_{X_2} and the finite orbits. This is what we saw on the torus a page ago, where two flows with only closed orbits combined into a walk with a unique stationary measure — except that here it is a theorem rather than a Fourier computation.*

The general theorem. As with Ratner’s theorems, the statement for X_2 is a special case of a general one, and the general one has the same three parts. Let G be a connected real Lie group, $\Lambda \leq G$ a lattice, $X = G/\Lambda$, and let μ be a compactly supported Borel probability measure on G . The hypothesis replacing Zariski density is now: the algebraic group generated by $\text{Ad}(\Gamma_\mu)$ is Zariski connected, semisimple, and has no compact factor.

Theorem 0.20 (Benoist–Quint). *With G, Λ, μ as above:*

- (i) (Stationary measures [BQ1].) *Every ergodic μ -stationary Borel probability measure on X is Γ_μ -invariant and homogeneous: it is the natural measure on a closed orbit $L \cdot x$ of a closed subgroup $L \leq G$ carrying a finite L -invariant measure.*
- (ii) (Orbit closures [BQ3, BQ4].) *For every $x \in X$ the closure $\overline{\Gamma_\mu \cdot x}$ is homogeneous: it equals $L \cdot x$ for such a subgroup L .*
- (iii) (Equidistribution [BQ2].) *For every $x \in X$, the Cesàro averages $\frac{1}{n} \sum_{k < n} \mu^{*k}$ converge weak-* to the natural measure on $\overline{\Gamma_\mu \cdot x}$, and almost every trajectory (ω_n) equidistributes towards it.*

Compare this with Theorem 0.9. The conclusions are word for word the same — measures and orbit closures are homogeneous, orbits equidistribute in their closures — and only the hypothesis has changed: instead of asking that the acting group be generated by unipotent one-parameter subgroups, one asks that the group generated by $\text{supp } \mu$ be large in the algebraic sense. Neither hypothesis requires the acting elements to be unipotent themselves, and the second does not even require the acting group to contain a unipotent element, which is what makes Remark 0.19 possible. Part (i) contains an assertion with no counterpart in Ratner’s theorem, sometimes called *stiffness*: on these spaces a stationary measure is automatically invariant. By Remark 0.17 this is a statement about X , not a formality.

Stiffness is also how the theorem is proved. One does not produce an invariant measure and compare; one shows that a stationary measure must be invariant, and only then appeals to the classification of invariant measures — so that Theorem 0.8 is an input to Theorem 0.18, which is one reason the three parts of this course come in the order they do. The route from stationarity to invariance runs through the theory of random walks on $\text{SL}_2(\mathbb{R})$: the Furstenberg measure of Remark 0.17, the Cartan decomposition of a random trajectory together with the

limit theorems describing it, and finally an *exponential drift* argument, modelled on Ratner’s polynomial one, which manufactures the missing invariance out of two nearby trajectories that the walk pushes apart.

0.8. Prerequisites and references. We assume the basics of measure theory, of topology, and enough Lie theory to be comfortable with matrix groups and their Lie algebras; the little hyperbolic geometry we need is recalled in §2.1, and the ergodic theory is introduced where it is used. Everything else is proved or precisely referenced.

The standard reference, and the one these notes follow most closely, is Einsiedler–Ward [EW3]; [EW1] covers the ergodic theory we use, and its Chapter 9 the hyperbolic geometry. Part I on Ratner’s theorems is inspired by the short book of Witte Morris [WM]. For Part III we follow the book of Benoist and Quint [BQ3] and the lecture notes of Bénard [Be], which give a self-contained account of Benoist–Quint’s classification of stationary measures [BQ1] in the case of $\mathrm{PSL}_2(\mathbb{R})$.

1. SOME BASIC VOCABULARY

This section fixes the language and the conventions that will be used throughout. It contains no dynamics, everything here is vocabulary. The dynamical notions — ergodicity, mixing, equidistribution — are introduced where they are first needed. For proofs and more details we refer to [EW3, Chapter 1].

1.1. Group actions, orbits and stabilisers.

Definition 1.1 (Action). Let G be a group and X a set. A *left action* of G on X is a map $G \times X \rightarrow X$, $(g, x) \mapsto g \cdot x$, such that

$$(1.1) \quad g \cdot (h \cdot x) = (gh) \cdot x, \quad e \cdot x = x \quad (g, h \in G, x \in X),$$

e being the identity of G . A *right action* is a map $X \times G \rightarrow X$, $(x, g) \mapsto x \cdot g$, with $(x \cdot g) \cdot h = x \cdot (gh)$ and $x \cdot e = x$.

In the case where G is a topological group and X a topological space, the action is called *continuous* if the defining map (1.1) is continuous.

Remark 1.2. A left action becomes a right action by setting $x \cdot g := g^{-1} \cdot x$, and conversely. The inverse is not optional: $(x \cdot g) \cdot h = x \cdot (gh)$ forces it.

For $x \in X$ the *orbit* of x is $G \cdot x := \{g \cdot x : g \in G\}$, and the *orbit closure* is $\overline{G \cdot x}$ (for a continuous action on a topological space). The *stabiliser* of $x \in X$ is $\text{Stab}_G(x) = \{g \in G : g \cdot x = x\}$, a subgroup of G ; it is closed (this uses that fact that the action is continuous and X is Hausdorff). The action is *transitive* if X is a single orbit, and *simply transitive* if moreover the stabilisers are trivial.

Definition 1.3 (Homogeneous space). A *homogeneous G -space* is a topological space X with a continuous transitive G -action.

If $H \leq G$ is a subgroup, the coset spaces

$$G/H = \{gH\}, \quad H \backslash G = \{Hg\}$$

carry the quotient topology. Here G acts on G/H on the left by $g_0 \cdot (gH) = (g_0g)H$, and on $H \backslash G$ on the right by $(Hg) \cdot g_0 = H(gg_0)$. The quotient map is always open. If H is closed, G/H and $H \backslash G$ are Hausdorff and locally compact. We will see that every homogeneous G -space is of this form.

Lemma 1.4 (Orbit–stabiliser). *The map*

$$(1.2) \quad G/\text{Stab}_G(x) \longrightarrow G \cdot x, \quad g \text{Stab}_G(x) \longmapsto g \cdot x,$$

is a well-defined G -equivariant bijection.

Proof. Write $S = \text{Stab}_G(x)$. The map is well defined and injective at once: for $g, h \in G$,

$$g \cdot x = h \cdot x \iff (h^{-1}g) \cdot x = x \iff h^{-1}g \in S \iff gS = hS,$$

where the first equivalence is the action axiom. It is surjective by the definition of the orbit, and equivariant because $g_0 \cdot (g \cdot x) = (g_0g) \cdot x$ is the image of g_0gS . $\square$

In the topological category the bijection (1.2) need not be a homeomorphism, but it is under hypotheses which will always be satisfied here.

Proposition 1.5. *Let G be a locally compact σ -compact topological group acting continuously and transitively on a locally compact Hausdorff space X . Then for every $x \in X$ the map of Lemma 1.4 is a homeomorphism $G/\text{Stab}_G(x) \rightarrow X$.*

In particular, every homogeneous G -space (with G, X as in Proposition 1.5) is of the form G/H where H is a closed subgroup. For a proof we refer to [EW3, Proposition 1.37]. The idea is that the map is a continuous bijection, so only openness is at issue: one covers G by countably many compacta, uses that X is a Baire space to find one whose image has non-empty interior, and translates.

Example 1.6. *Transitivity is essential in Proposition 1.5. Let $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ and let $\mathbb{R}$ act on the torus $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$ by $t \cdot (x, y) = (x + t, y + \alpha t)$. All stabilisers are trivial, so Lemma 1.4 gives a bijection $\mathbb{R} \rightarrow \mathbb{R} \cdot (0, 0)$; but the orbit is dense in $\mathbb{T}^2$, so with the subspace topology it is not homeomorphic to $\mathbb{R}$. Orbits are in general not embedded submanifolds, and this is one of the reasons orbit closures are interesting.*

1.2. The space of lattices. A lattice in $\mathbb{R}^d$ is a discrete subgroup $\Lambda \subseteq \mathbb{R}^d$ which spans $\mathbb{R}^d$ as a vector space; equivalently,

$$\Lambda = g\mathbb{Z}^d = \mathbb{Z}ge_1 \oplus \cdots \oplus \mathbb{Z}ge_d$$

for some $g \in \mathrm{GL}_d(\mathbb{R})$. The quantity $\mathrm{covol}(\Lambda) := |\det g|$ does not depend on the choice of g , since two such matrices differ by an element of $\mathrm{GL}_d(\mathbb{Z})$, whose determinant is ± 1 ; geometrically it is the volume of the parallelepiped spanned by any basis, i.e. of a fundamental domain for Λ acting on $\mathbb{R}^d$ by translations. Call Λ *unimodular* if $\mathrm{covol}(\Lambda) = 1$, and write $\mathcal{L}_d$ for the set of unimodular lattices in $\mathbb{R}^d$.

The group $G = \mathrm{SL}_d(\mathbb{R})$ acts on $\mathcal{L}_d$ on the left, by the linear action $g \cdot \Lambda := g\Lambda$; this preserves $\mathcal{L}_d$ because $\mathrm{covol}(g\Lambda) = |\det g| \mathrm{covol}(\Lambda)$.

The action is transitive. Let $\Lambda = h\mathbb{Z}^d \in \mathcal{L}_d$ with $h \in \mathrm{GL}_d(\mathbb{R})$, so $|\det h| = 1$. If $\det h = -1$, replace h by $h \mathrm{diag}(-1, 1, \dots, 1)$, which changes neither $h\mathbb{Z}^d$ nor the absolute value of the determinant. So we may take $h \in \mathrm{SL}_d(\mathbb{R})$, and then $\Lambda = h \cdot \mathbb{Z}^d$.

The stabiliser of $\mathbb{Z}^d$ is $\mathrm{SL}_d(\mathbb{Z})$. If $g\mathbb{Z}^d = \mathbb{Z}^d$ then g maps the standard basis to integer vectors, so g has integer entries; the same applies to g^{-1} ; hence $g \in \mathrm{GL}_d(\mathbb{Z})$, and $\det g = 1$ gives $g \in \mathrm{SL}_d(\mathbb{Z})$. The converse is clear.

By Proposition 1.5 we therefore have a homeomorphism

$$(1.3) \quad X_d := \mathrm{SL}_d(\mathbb{R})/\mathrm{SL}_d(\mathbb{Z}) \longrightarrow \mathcal{L}_d, \quad g \mathrm{SL}_d(\mathbb{Z}) \longmapsto g\mathbb{Z}^d,$$

and it is equivariant: the left action of $\mathrm{SL}_d(\mathbb{R})$ on the coset space corresponds to the linear action on lattices. By Proposition 1.5 it is a homeomorphism once $\mathcal{L}_d$ is given its natural topology ([EW1, Ch. 11]).

1.3. Distance and injectivity radius. From now on until the end of this section, the standing assumption is that G is a locally compact σ -compact topological group.

Definition 1.7 (Discrete subgroup). A subgroup $\Gamma \leq G$ of a topological group is *discrete* if the subspace topology on Γ is discrete; equivalently, if there is a neighbourhood $U \ni e$ in G with $U \cap \Gamma = \{e\}$.

Homogeneous dynamics studies the spaces $X = G/\Gamma$, where $\Gamma \leq G$ is a discrete subgroup, and the action on X of a closed subgroup $H \leq G$. We shall want to say that two points of X are close.

Lemma 1.8 (The metric on G/Γ). *Let G be a locally compact group carrying a right-invariant metric d_G inducing its topology (for a Lie group, take the distance of a right-invariant Riemannian metric), and let $\Gamma \leq G$ be discrete. Then*

$$d_X(g\Gamma, h\Gamma) := \inf_{\gamma \in \Gamma} d_G(g, h\gamma)$$

is a well-defined metric on $X = G/\Gamma$ inducing the quotient topology, and

$$d_X(x, g \cdot x) \leq d_G(e, g) \quad \text{for all } x \in X, g \in G.$$

Proof. Well-definedness and the triangle inequality follow from right invariance; positivity uses that $h\Gamma$ is closed and discrete (Lemma 1.11). For the estimate write $x = h\Gamma$ and use $d_X(h\Gamma, gh\Gamma) \leq d_G(h, gh) = d_G(e, g)$. $\square$

The point of the last inequality is that it is *uniform in x* : moving a point of X by a group element close to the identity moves it a small distance, no matter where it is.

Definition 1.9 (Injectivity radius). For $x \in X = G/\Gamma$ let

$$\text{inj}(x) := \sup\{r > 0 : g \mapsto g \cdot x \text{ is injective on } B_r^G(e)\}.$$

The supremum is over a non-empty set because Γ is discrete; thus $\text{inj}(x) > 0$ for every x .

On a non-compact X the function inj is in general not bounded away from 0 — this is exactly what “going into the cusp” will mean. On the space of lattices the failure is detected by a single number.

Example 1.10 (The first minimum). For a lattice $\Lambda \subseteq \mathbb{R}^d$ write

$$\lambda_1(\Lambda) := \min_{v \in \Lambda \setminus \{0\}} \|v\|$$

for the length of a shortest non-zero vector; the minimum is attained because Λ is discrete. Under the dictionary (1.3), λ_1 becomes a continuous function on X_d , and small injectivity radius and small first minimum are two names for the same degeneration. The qualitative statement is: a family of unimodular lattices has injectivity radius bounded away from 0 if and only if its first minima are bounded away from 0.

1.4. Fundamental domains and lattices.

Lemma 1.11. Let G be a locally compact Hausdorff group and $\Gamma \leq G$ discrete. Then

- (a) Γ is closed in G ;
- (b) Γ acts properly discontinuously on G by right translation: for every compact $K \subseteq G$ the set $\{\gamma \in \Gamma : K\gamma \cap K \neq \emptyset\}$ is finite;
- (c) consequently $X = G/\Gamma$ is locally compact Hausdorff and the projection $\pi : G \rightarrow X$ is a local homeomorphism.

Proof. For (b), note that $K\gamma \cap K \neq \emptyset$ forces $\gamma \in K^{-1}K$, which is compact; a discrete closed subset of a compact set is finite. Part (a) is the standard fact that a discrete subgroup of a Hausdorff group is closed, and (c) follows from (b). See [EW1, Ch. 9]. $\square$

Definition 1.12 (Fundamental domain). A (strict, Borel) fundamental domain for the right action of Γ on G is a Borel set $F \subseteq G$ such that $F\Gamma = G$ and $F\gamma \cap F = \emptyset$ for every $\gamma \in \Gamma \setminus \{e\}$; equivalently, $|F \cap g\Gamma| = 1$ for every $g \in G$.

Such an F always exists when G is second countable:

Lemma 1.13 (Fundamental domain). Let $\Gamma \leq G$ be a discrete subgroup. Then there exists a fundamental domain $F \subseteq G$ for Γ .

For a proof we refer the reader to [EW3, Lemma 1.10]. In practice one produces a closed set E with $E\Gamma = G$ whose interior meets each orbit at most once, and then repairs the boundary.

1.5. Haar measure and the measure on G/Γ .

Definition 1.14 (Haar measure). A *left Haar measure* on a locally compact Hausdorff group G is a non-zero Radon measure m_G with $m_G(gA) = m_G(A)$ for all Borel A and all $g \in G$. It exists and is unique up to a positive scalar.

For existence and uniqueness see [F, Ch. 11] or [EW3, Appendix A].

Definition 1.15 (Lattice). Let G be a locally compact group with left Haar measure m_G (§1.5) and let $\Gamma \leq G$ be discrete with fundamental domain F . The *covolume* of Γ is $\text{vol}(G/\Gamma) := m_G(F)$; it does not depend on the choice of F (Proposition 1.17). If it is finite, Γ is called a *lattice*. A lattice is *uniform* (or *cocompact*) if G/Γ is compact, and *non-uniform* otherwise.

Example 1.16. We will see later in the course that $\Gamma = \text{SL}_d(\mathbb{Z})$ is a non-uniform lattice in $G = \text{SL}_d(\mathbb{R})$.

A left Haar measure need not be right invariant; the defect is the *modular function* $\Delta_G : G \rightarrow \mathbb{R}_{>0}$, a continuous homomorphism determined by $m_G(Ag) = \Delta_G(g) m_G(A)$. The group is *unimodular* if $\Delta_G \equiv 1$. Abelian, compact, discrete and semisimple groups are unimodular; in particular $\text{SL}_d(\mathbb{R})$ is (proved in Chapter 2). The subgroup $B = U^-A$ of upper triangular matrices is *not*, which is why a right Haar measure appears in Lemma 3.1.

Proposition 1.17 (Unfolding). Let G be unimodular and $\Gamma \leq G$ discrete. There is a unique G -invariant Radon measure m_X on $X = G/\Gamma$ such that

$$(1.4) \quad \int_G f dm_G = \int_X \left(\sum_{\gamma \in \Gamma} f(g\gamma) \right) dm_X(g\Gamma) \quad (f \in C_c(G)),$$

and $m_X(X) = m_G(F)$ for any fundamental domain F . In particular Γ is a lattice if and only if X carries a finite G -invariant measure, which one then normalises to a probability measure μ_X .

Proof. Fix a Borel fundamental domain F and let $\pi : G \rightarrow X$ be the projection. By Definition 1.12 the restriction $\pi|_F$ is a Borel bijection onto X , so we may define

$$m_X := (\pi|_F)_*(m_G|_F), \quad \text{that is} \quad m_X(A) = m_G(F \cap \pi^{-1}A).$$

Independence of F . Let F' be another fundamental domain. Since $\pi^{-1}A$ is invariant under right translation by Γ and m_G is also right invariant (G is unimodular),

$$m_G(F \cap \pi^{-1}A) = \sum_{\gamma \in \Gamma} m_G(F \cap F'\gamma \cap \pi^{-1}A) = \sum_{\gamma \in \Gamma} m_G(F\gamma^{-1} \cap F' \cap \pi^{-1}A) = m_G(F' \cap \pi^{-1}A),$$

using $G = \bigsqcup_{\gamma} F'\gamma$ for the first equality and $G = \bigsqcup_{\gamma} F\gamma^{-1}$ for the last.

Invariance. For $g_0 \in G$ the set g_0F is again a fundamental domain, and by left invariance of m_G ,

$$m_X(g_0A) = m_G(F \cap \pi^{-1}(g_0A)) = m_G(g_0(g_0^{-1}F \cap \pi^{-1}A)) = m_G(g_0^{-1}F \cap \pi^{-1}A) = m_X(A),$$

the last step by independence of the fundamental domain.

The unfolding formula. For $f \in C_c(G)$, using $G = \bigsqcup_{\gamma} F\gamma$ and right invariance of m_G ,

$$\int_G f dm_G = \sum_{\gamma \in \Gamma} \int_{F\gamma} f dm_G = \sum_{\gamma \in \Gamma} \int_F f(g\gamma) dm_G(g) = \int_F \sum_{\gamma \in \Gamma} f(g\gamma) dm_G(g),$$

the interchange being legitimate because all terms are non-negative for $f \geq 0$, and the general case follows by linearity. The right-hand side is $\int_X (\sum_{\gamma} f(g\gamma)) dm_X$ by the definition of m_X . Taking $A = X$ gives $m_X(X) = m_G(F)$. Uniqueness holds because every element of $C_c(X)$ is of the form $g\Gamma \mapsto \sum_{\gamma} f(g\gamma)$ for some $f \in C_c(G)$, so (1.4) determines m_X on $C_c(X)$; see [EW3, Ch. 1]. $\square$

Remark 1.18. *Unimodularity is not a technicality one may drop: for $H \leq G$ closed, an invariant Radon measure on G/H exists if and only if $\Delta_G|_H = \Delta_H$. For discrete $H = \Gamma$ this says $\Delta_G|_{\Gamma} \equiv 1$, and if G admits a lattice then G is unimodular. So the hypothesis above is essentially forced.*

1.6. A vocabulary for orbits. We can now name the objects the course is about. Throughout, $X = G/\Gamma$ with Γ a lattice, $H \leq G$ is a closed subgroup, and $x = g\Gamma \in X$. The orbit is $Hx = \{hg\Gamma : h \in H\}$ and its *orbit closure* is $\overline{Hx}$. The stabiliser of x in H is

$$\text{Stab}_H(x) := \{h \in H : hx = x\} = H \cap g\Gamma g^{-1},$$

a discrete subgroup of H ; by Lemma 1.4 the orbit is a continuous bijective image of $H/\text{Stab}_H(x)$.

The following four adjectives describe everything we shall want to say about a single orbit.

Definition 1.19 (Types of orbits). The orbit Hx is called

- (i) *closed* if it is closed in X , and *compact* if it is a compact subset of X ;
- (ii) *periodic*, when $H = \{g_t\}_{t \in \mathbb{R}}$ is a one-parameter subgroup, if $\text{Stab}_H(x) = \ell\mathbb{Z}$ for some $\ell > 0$; the number ℓ is then the *length* of the orbit;
- (iii) *divergent* if the orbit map $H \rightarrow X, h \mapsto hx$, is proper, i.e. if $\{h \in H : hx \in K\}$ is relatively compact in H for every compact $K \subseteq X$ — informally, the orbit eventually leaves every compact set;
- (iv) *dense* if $\overline{Hx} = X$.

For a one-parameter subgroup, “compact” and “periodic” coincide, so the two words may be used interchangeably; this is proved in Lemma 2.7. A finer notion, which requires the measure μ_X , is the following: the orbit of x under $\{g_t\}$ is *equidistributed* if

$$\frac{1}{T} \int_0^T f(g_t x) dt \longrightarrow \int_X f d\mu_X \quad (T \rightarrow \infty)$$

for every $f \in C_c(X)$; equivalently, the normalised orbit measures converge to μ_X in the weak-* topology. Equidistribution implies density, and is strictly stronger.

The next lemma explains why, in practice, closed orbits and finite volume go together; we shall use it without further comment.

Lemma 1.20 (Finite volume orbits are closed). *If Σ_x is a lattice in H — equivalently, if the orbit Hx carries a finite H -invariant measure — then Hx is closed in X .*

For the proof see [EW3, Ch. 1]. The converse is false in general: an orbit can be closed and of infinite volume, as the divergent A -orbits of §2.4 will show.

With this vocabulary the questions of the course can be stated.

- (1) *Orbit closures.* What is $\overline{Hx}$? When is the orbit dense?
- (2) *Closed orbits.* Which points have closed, compact or divergent orbits, and what do those points look like?
- (3) *Recurrence and escape of mass.* Does the orbit return to a fixed compact set, or does it spend more and more time near infinity?
- (4) *Invariant measures and equidistribution.* Which H -invariant probability measures are there, and do long orbit pieces distribute according to one of them?
- (5) *Rigidity and arithmetic.* Are orbit closures and invariant measures forced to be algebraic, and what arithmetic information do they carry?

The recurring theme of these notes, visible already in Chapter 2, is the correspondence

$$\text{closed orbit} \longleftrightarrow \text{arithmetic object},$$

together with the observation that it is subtle: it holds for the diagonal subgroup A and for the horospherical subgroup U^- , and fails completely for $K = \mathrm{SO}_2(\mathbb{R})$ and for $B = U^-A$.

A dictionary.

algebra	dynamics
G/Γ	phase space
$H \leq G$	the acting group
$Hg\Gamma$	orbit of $x = g\Gamma$
$\overline{Hg\Gamma}$	orbit closure
$H \cap g\Gamma g^{-1}$	periods of the orbit
Haar measure	natural invariant measure
Γ a lattice	finite-volume quotient
Γ non-uniform	cusps, escape of mass
algebraic subgroup	rigid dynamical behaviour

2. THE DYNAMICAL PICTURE OF $\mathrm{SL}_2(\mathbb{Z}) \backslash \mathrm{SL}_2(\mathbb{R})$

We now turn to the smallest example that already exhibits most of the dynamical phenomena we shall be interested in, namely the quotient

$$X_2 = \mathrm{SL}_2(\mathbb{Z}) \backslash \mathrm{SL}_2(\mathbb{R}).$$

The point of this section is to have a concrete picture to hold on to. In fact we describe two geometric ways of thinking about X_2 : (1) as the unit tangent bundle of the modular surface, and (2) as the space of unimodular lattices in $\mathbb{R}^2$. We then classify some of the orbit behaviours of the diagonal and unipotent subgroups of $\mathrm{SL}_2(\mathbb{R})$.

2.1. The upper half-plane and its isometries. Recall that the *upper half-plane* is

$$\mathbb{H} = \{z = x + iy \in \mathbb{C} : y = \mathrm{Im}(z) > 0\},$$

equipped with the Riemannian metric

$$(2.1) \quad \langle u, v \rangle_z = \frac{u \cdot v}{y^2}, \quad (z, u), (z, v) \in T_z \mathbb{H} = \{z\} \times \mathbb{C},$$

where $u \cdot v$ is the Euclidean inner product after identifying $u, v \in \mathbb{C}$ with elements of $\mathbb{R}^2$. Thus at each point the hyperbolic inner product is a positive multiple of the Euclidean one — one says that the metric (2.1) is *conformal* to the Euclidean metric — so that hyperbolic and Euclidean angles agree, although lengths do not. The associated length of a piecewise smooth curve $\gamma: [0, 1] \rightarrow \mathbb{H}$ and the associated distance are

$$\ell(\gamma) = \int_0^1 \frac{|\gamma'(s)|}{\mathrm{Im} \gamma(s)} ds, \quad d_{\mathbb{H}}(z, w) = \inf \{\ell(\gamma) : \gamma \text{ from } z \text{ to } w\}.$$

This is the upper half-plane model of the hyperbolic plane; see for instance [EW1, Ch. 9].

The group $\mathrm{SL}_2(\mathbb{R})$ acts on $\mathbb{H}$ by Möbius transformations,

$$(2.2) \quad g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : \quad \mathbb{H} \ni z \mapsto g \cdot z = \frac{az + b}{cz + d}.$$

Lemma 2.1. *The formula (2.2) defines a transitive action of $\mathrm{SL}_2(\mathbb{R})$ on $\mathbb{H}$ by isometries, and $\mathrm{Stab}_{\mathrm{SL}_2(\mathbb{R})}(i) = \mathrm{SO}_2(\mathbb{R})$. Consequently*

$$\mathrm{SL}_2(\mathbb{R}) / \mathrm{SO}_2(\mathbb{R}) \cong \mathbb{H}, \quad g \mathrm{SO}_2(\mathbb{R}) \mapsto g \cdot i.$$

Proof. The formula lands in $\mathbb{H}$. For g as in (2.2) and $z = x + iy$,

$$(2.3) \quad \mathrm{Im}(g \cdot z) = \frac{\mathrm{Im}((az + b)\overline{(cz + d)})}{|cz + d|^2} = \frac{(ad - bc)y}{|cz + d|^2} = \frac{y}{|cz + d|^2} > 0,$$

and $cz + d \neq 0$ because $z \notin \mathbb{R}$ and $(c, d) \neq (0, 0)$.

It is an action. Introduce the *automorphy factor*

$$j(g, z) := cz + d,$$

and observe that (2.2) is the projective action of g on the column vector $\begin{pmatrix} z \\ 1 \end{pmatrix}$:

$$(2.4) \quad g \begin{pmatrix} z \\ 1 \end{pmatrix} = \begin{pmatrix} az + b \\ cz + d \end{pmatrix} = j(g, z) \begin{pmatrix} g \cdot z \\ 1 \end{pmatrix}.$$

Applying (2.4) twice, for $g, h \in \mathrm{SL}_2(\mathbb{R})$,

$$(gh) \begin{pmatrix} z \\ 1 \end{pmatrix} = g \left(j(h, z) \begin{pmatrix} h \cdot z \\ 1 \end{pmatrix} \right) = j(h, z) j(g, h \cdot z) \begin{pmatrix} g \cdot (h \cdot z) \\ 1 \end{pmatrix},$$

and comparing with (2.4) for gh gives both

$$(2.5) \quad g \cdot (h \cdot z) = (gh) \cdot z \quad \text{and} \quad j(gh, z) = j(g, h \cdot z) j(h, z).$$

Since $I \cdot z = z$, this is an action.

The induced action on tangent vectors. The map (2.2) is holomorphic in z with complex derivative

$$\frac{d}{dz} \frac{az + b}{cz + d} = \frac{a(cz + d) - c(az + b)}{(cz + d)^2} = \frac{ad - bc}{(cz + d)^2} = j(g, z)^{-2}.$$

The differential at z of a holomorphic map is $\mathbb{C}$ -linear, namely multiplication by that derivative; identifying $T_z \mathbb{H}$ with $\mathbb{C}$ we therefore get

$$(2.6) \quad Dg: (z, v) \mapsto (g \cdot z, j(g, z)^{-2} v).$$

This is again an action, now on $T\mathbb{H} = \mathbb{H} \times \mathbb{C}$. Indeed $DI = \mathrm{id}$, and for $g, h \in \mathrm{SL}_2(\mathbb{R})$ we may compose (2.6) with itself:

$$Dg(Dh(z, v)) = Dg(h \cdot z, j(h, z)^{-2} v) = (g \cdot (h \cdot z), j(g, h \cdot z)^{-2} j(h, z)^{-2} v).$$

By the two identities in (2.5) the first coordinate is $(gh) \cdot z$ and the second is

$$(j(g, h \cdot z) j(h, z))^{-2} v = j(gh, z)^{-2} v,$$

so that $Dg \circ Dh = D(gh)$, which is precisely the chain rule for the composition of the two Möbius maps. The cocycle relation in (2.5) is the chain rule, written for derivatives instead of for the maps themselves.

Each g is conformal. Call a smooth injective map φ between open subsets of $\mathbb{C}$ *conformal* if its differential preserves angles at every point. By (2.6) the differential of $z \mapsto g \cdot z$ is multiplication by the non-zero complex number $j(g, z)^{-2}$, that is, a rotation followed by a dilation; such a map preserves angles, so g acts conformally. We shall use this in Lemma 2.4.

The action is isometric. Recall that a diffeomorphism φ of a Riemannian manifold is an *isometry* if

$$\langle D\varphi(u), D\varphi(v) \rangle_{\varphi(z)} = \langle u, v \rangle_z$$

for all tangent vectors u, v at all points z ; an isometry preserves the lengths of curves, and hence the distance $d_{\mathbb{H}}$. To verify this condition it is enough to check that $D\varphi$ preserves *norms*,

$$(2.7) \quad \|D\varphi(v)\|_{\varphi(z)} = \|v\|_z \quad \text{for all } v,$$

because $D\varphi$ is linear and an inner product is determined by its norm through the polarisation identity $\langle u, v \rangle = \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2)$.

So let us verify (2.7) for $\varphi: z \mapsto g \cdot z$. By (2.6) the differential at z is multiplication by $j(g, z)^{-2}$, which scales Euclidean norms by $|j(g, z)|^{-2}$; and by (2.3) the denominator in (2.1) changes by the compensating factor, since $\mathrm{Im}(g \cdot z) = \mathrm{Im}(z) |j(g, z)|^{-2}$. Hence

$$\|Dg(z, v)\|_{g \cdot z} = \frac{|j(g, z)|^{-2} |v|}{\mathrm{Im}(g \cdot z)} = \frac{|j(g, z)|^{-2} |v|}{\mathrm{Im}(z) |j(g, z)|^{-2}} = \frac{|v|}{y} = \|v\|_z.$$

Transitivity. Writing

$$(2.8) \quad b(x, y) := \begin{pmatrix} 1 & x \\ & 1 \end{pmatrix} \begin{pmatrix} y^{1/2} & \\ & y^{-1/2} \end{pmatrix} = \begin{pmatrix} y^{1/2} & xy^{-1/2} \\ & y^{-1/2} \end{pmatrix} \quad (x \in \mathbb{R}, y > 0),$$

one computes $b(x, y) \cdot i = x + iy$, so every point of $\mathbb{H}$ lies in the orbit of i .

The stabiliser of i . $g \cdot i = i$ means $ai + b = i(ci + d) = -c + di$, that is $a = d$ and $b = -c$; together with $ad - bc = a^2 + b^2 = 1$ this says precisely that $g \in \mathrm{SO}_2(\mathbb{R})$. The last assertion is now Lemma 1.3. $\square$

Since g acts isometrically, the action (2.6) preserves the *unit tangent bundle*

$$T^1\mathbb{H} = \{(z, v) \in T\mathbb{H} : \|v\|_z = 1\}.$$

The kernel of the action on $\mathbb{H}$ is $\{\pm I\}$: if $g \cdot z = z$ for every z then $cz^2 + (d-a)z - b = 0$ for every $z \in \mathbb{H}$, and a non-zero polynomial has finitely many roots, so $b = c = 0$ and $a = d = \pm 1$.

Lemma 2.2. *The action of $\mathrm{PSL}_2(\mathbb{R}) = \mathrm{SL}_2(\mathbb{R})/\{\pm I\}$ on $T^1\mathbb{H}$ is simply transitive. In particular*

$$\mathrm{PSL}_2(\mathbb{R}) \cong T^1\mathbb{H}, \quad g \longmapsto Dg(i, \uparrow),$$

where $\uparrow$ denotes the upward pointing vector of hyperbolic length 1 at a given point of $\mathbb{H}$. The identification is equivariant for the left action of $\mathrm{PSL}_2(\mathbb{R})$ on both sides.

Proof. At the point i we have $\uparrow = i \in \mathbb{C} = T_i\mathbb{H}$, since $\|i\|_i = |i|/1 = 1$. For $k_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \in \mathrm{SO}_2(\mathbb{R})$ we have $j(k_\theta, i) = \sin \theta \cdot i + \cos \theta = e^{i\theta}$, so by (2.6)

$$(2.9) \quad Dk_\theta(i, v) = (i, e^{-2i\theta}v) :$$

the stabiliser $\mathrm{SO}_2(\mathbb{R})$ of i rotates $T_i^1\mathbb{H}$ at double speed. Hence $\mathrm{SO}_2(\mathbb{R})$ acts transitively on the circle $T_i^1\mathbb{H}$, which together with Lemma 2.1 gives transitivity on $T^1\mathbb{H}$. By (2.9) an element k_θ fixing $(i, \uparrow)$ satisfies $e^{-2i\theta} = 1$, i.e. $\theta \in \pi\mathbb{Z}$, i.e. $k_\theta = \pm I$. So the stabiliser of $(i, \uparrow)$ in $\mathrm{PSL}_2(\mathbb{R})$ is trivial. Equivariance is $D(g_0g) = Dg_0 \circ Dg$, proved in Lemma 2.1. $\square$

In particular, if $\Gamma < \mathrm{PSL}_2(\mathbb{R})$ is a discrete subgroup, then Γ acts on $T^1\mathbb{H}$ on the left, and Lemma 2.2 identifies

$$T^1(\Gamma \backslash \mathbb{H}) \cong \Gamma \backslash \mathrm{PSL}_2(\mathbb{R}),$$

which is the reason for writing the quotient on the left in this section.

Recall the definition of a *geodesic*:

Definition 2.3 (Geodesic). Let $(M, \langle \cdot, \cdot \rangle)$ be a Riemannian manifold with induced distance d . A smooth curve $\gamma: I \rightarrow M$ is a *geodesic* if it has constant speed, $\|\gamma'(t)\|_{\gamma(t)} \equiv c$, and is locally length-minimising: every $t_0 \in I$ has a neighbourhood $[a, b] \subseteq I$ with $\ell(\gamma|_{[a, b]}) = d(\gamma(a), \gamma(b))$. It is *unit-speed* if $c = 1$.

We shall use the following classical description of the geodesics of $\mathbb{H}$; we include the short argument because we need the precise normalisation in §2.3.

Lemma 2.4. *The imaginary axis is a geodesic, parametrised with unit speed by $t \mapsto e^ti$; in particular $d_{\mathbb{H}}(i, e^ti) = |t|$. Every geodesic of $\mathbb{H}$ is the image of the imaginary axis under some element of $\mathrm{SL}_2(\mathbb{R})$, hence is a vertical half-line or a half-circle meeting $\mathbb{R} = \partial\mathbb{H}$ orthogonally.*

Proof. Let $\gamma(s) = x(s) + iy(s)$, $s \in [0, 1]$, be a piecewise smooth path from ia to ib with $0 < a < b$. Its hyperbolic length is

$$\int_0^1 \frac{\sqrt{x'(s)^2 + y'(s)^2}}{y(s)} ds \geq \int_0^1 \frac{|y'(s)|}{y(s)} ds \geq \left| \int_0^1 \frac{y'(s)}{y(s)} ds \right| = \log \frac{b}{a},$$

with equality if and only if $x' \equiv 0$ and y is monotone. Hence the segment of the imaginary axis is the unique shortest path, and $d_{\mathbb{H}}(ia, ib) = \log(b/a)$; with $a = 1$, $b = e^t$ this is the first claim, and the parametrisation $t \mapsto e^t i$ has unit speed because $\|\frac{d}{dt} e^t i\|_{e^t i} = e^t / e^t = 1$.

For the second claim, $\mathrm{SL}_2(\mathbb{R})$ acts by isometries and, by Lemma 2.2, transitively on unit tangent vectors; since a geodesic is determined by its initial position and direction, every geodesic is the image of the imaginary axis. Finally, the image of the imaginary axis under (2.2) is a vertical half-line (if $c = 0$) or a half-circle centred on $\mathbb{R}$ (if $c \neq 0$), as one sees by extending (2.2) to $\mathbb{C} \cup \{\infty\}$, where Möbius maps send circles and lines to circles and lines. It meets $\mathbb{R}$ orthogonally because g is conformal, as observed in the proof of Lemma 2.1: the imaginary axis meets $\mathbb{R}$ at a right angle, and a conformal map preserves that angle. $\square$

2.2. The modular group and a fundamental domain. For dynamical purposes $\mathbb{H}$ is too big: $\mathrm{SL}_2(\mathbb{R})$ acts on it transitively. We therefore fold it up by a discrete subgroup $\Gamma < \mathrm{PSL}_2(\mathbb{R})$. Ideally one would like the quotient to be compact, but this is too restrictive in practice, and the most interesting example is not.

Recall from Definition 1.12 that a *fundamental domain* for a group Γ acting on a space Y is a Borel set $E \subseteq Y$ meeting each orbit exactly once: for every $g \in G$, $|E \cap \Gamma g| = 1$. Equivalently,

$$G = \bigsqcup_{\gamma \in \Gamma} \gamma E.$$

Proposition 2.5 (Fundamental domain). *Let $\Gamma = \mathrm{PSL}_2(\mathbb{Z}) = \mathrm{SL}_2(\mathbb{Z})/\{\pm I\}$ and*

$$E = \left\{ z \in \mathbb{H} : |\mathrm{Re} z| \leq \frac{1}{2}, |z| \geq 1 \right\}.$$

Then $\Gamma \cdot E = \mathbb{H}$, and for z in the interior of E the only $\gamma \in \Gamma$ with $\gamma \cdot z \in E$ is $\gamma = e$. After a harmless modification of the boundary, E is a fundamental domain for the action of Γ on $\mathbb{H}$.

Proof. Write $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, so that $T \cdot z = z + 1$ and $S \cdot z = -1/z$.

Every orbit meets E . Fix $z \in \mathbb{H}$. For $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$ we have $\mathrm{Im}(\gamma \cdot z) = y/|cz + d|^2$ by (2.3). As (c, d) runs over the non-zero integer vectors, $cz + d$ runs over the non-zero elements of the lattice $\mathbb{Z}z + \mathbb{Z}$ of $\mathbb{C}$, which is discrete; hence $|cz + d|$ has a positive minimum, and $\mathrm{Im}(\gamma \cdot z)$ attains a maximum over $\gamma \in \Gamma$, say at γ_0 . Replacing γ_0 by $T^n \gamma_0$ changes neither this imaginary part nor the maximality, and for a suitable n it achieves $|\mathrm{Re}(\gamma_0 \cdot z)| \leq \frac{1}{2}$. Put $w = \gamma_0 \cdot z$. If we had $|w| < 1$ then

$$\mathrm{Im}(S \cdot w) = \frac{\mathrm{Im} w}{|w|^2} > \mathrm{Im} w,$$

contradicting maximality. Hence $|w| \geq 1$ and $w \in E$.

Uniqueness in the interior. Suppose z and $\gamma \cdot z$ both lie in E with $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \neq e$, and assume without loss of generality that $\mathrm{Im}(\gamma \cdot z) \geq \mathrm{Im} z$, i.e. $|cz + d| \leq 1$. Since $z \in E$ we have $\mathrm{Im} z \geq \frac{\sqrt{3}}{2}$, so

$$1 \geq |cz + d| \geq |c| \mathrm{Im} z \geq |c| \frac{\sqrt{3}}{2},$$

giving $|c| \leq \frac{2}{\sqrt{3}} < 2$, so $c \in \{0, \pm 1\}$.

If $c = 0$ then $ad = 1$ with $a, d \in \mathbb{Z}$, so $\gamma = \pm T^n$ acts as $z \mapsto z + n$ with $n \neq 0$; but then $|\operatorname{Re} z|$ and $|\operatorname{Re} z + n|$ cannot both be $\leq \frac{1}{2}$ unless $|\operatorname{Re} z| = \frac{1}{2}$, i.e. $z \in \partial E$.

If $|c| = 1$ we may take $c = 1$ (replace γ by $-\gamma$), so $|z + d| \leq 1$. For $z \in E$ and $d \in \mathbb{Z}$,

$$|z + d|^2 = |z|^2 + 2d \operatorname{Re} z + d^2 \geq 1 - |d| + d^2,$$

which is > 1 for $|d| \geq 2$; for $d = 0$ it gives $|z| \leq 1$, hence $|z| = 1$, and for $|d| = 1$ equality forces $|z| = 1$ and $\operatorname{Re} z = \mp \frac{1}{2}$. In all cases $z \in \partial E$.

So for z in the interior of E the only possibility is $\gamma = e$. $\square$

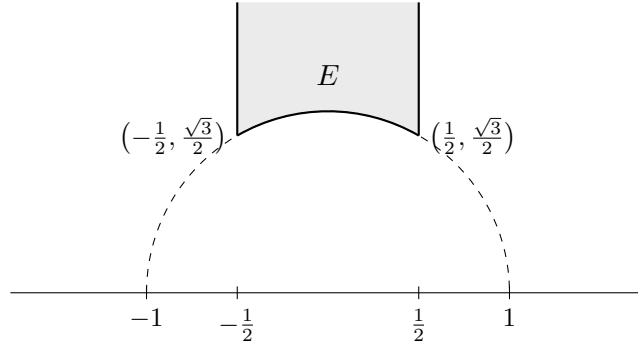


FIGURE 5. A fundamental domain $E \subseteq \mathbb{H}$ for the action of $\operatorname{PSL}_2(\mathbb{Z})$.

By Lemma 2.2 this also gives a fundamental domain upstairs in the group. Put

$$(2.10) \quad F := \{g \in \operatorname{PSL}_2(\mathbb{R}) : g \cdot i \in E\}.$$

Lemma 2.6. *F is a fundamental domain for the left action of Γ on $\operatorname{PSL}_2(\mathbb{R})$, up to a set of measure zero.*

Proof. Fix $g \in \operatorname{PSL}_2(\mathbb{R})$ and put $w = g \cdot i$. For $\gamma \in \Gamma$,

$$\gamma g \in F \iff (\gamma g) \cdot i \in E \iff \gamma \cdot w \in E,$$

using that the action is an action. By Proposition 2.5 the orbit $\Gamma \cdot w$ meets E in exactly one point, so the set of admissible γ is a single coset of $\operatorname{Stab}_\Gamma(w)$. That stabiliser is trivial except for the countably many w in the orbits of i and of $\frac{1}{2} + \frac{\sqrt{3}}{2}i$, which form a null set. Hence $|F \cap \Gamma g| = 1$ for almost every g , as required. $\square$

2.3. X_2 has finite volume. We now show that $\operatorname{SL}_2(\mathbb{Z})$ is a lattice in $\operatorname{SL}_2(\mathbb{R})$. Three facts are needed, and we prove them all for $\operatorname{SL}_2(\mathbb{R})$.

Lemma 2.7 (Unimodularity). *$\operatorname{SL}_2(\mathbb{R})$ is unimodular: a left Haar measure is also right invariant.*

Proof. The modular function $\Delta: \operatorname{SL}_2(\mathbb{R}) \rightarrow \mathbb{R}_{>0}$ is a continuous homomorphism into an abelian group. For $s \neq 0$ write $u_s = \begin{pmatrix} 1 & s \\ & 1 \end{pmatrix}$ and note

$$\begin{pmatrix} t & \\ & t^{-1} \end{pmatrix} u_s \begin{pmatrix} t^{-1} & \\ & t \end{pmatrix} = u_{t^2 s} \quad (t > 0),$$

so all u_s with $s > 0$ are conjugate; and Δ , being a homomorphism into an abelian group, is constant on conjugacy classes. Write $c := \Delta(u_s)$ for $s > 0$. Then $u_s u_{s'} = u_{s+s'}$ gives $c \cdot c = c$, hence $c = 1$, so $\Delta \equiv 1$ on $\{u_s\}$ and likewise on

the lower triangular unipotents $u_s^- = \begin{pmatrix} 1 & \\ & s \end{pmatrix}$. These generate $\mathrm{SL}_2(\mathbb{R})$ — every matrix of determinant 1 is a product of elementary matrices — so $\Delta \equiv 1$. $\square$

Lemma 2.8 (Iwasawa decomposition). *Let $N = \{u_x : x \in \mathbb{R}\}$, $A = \{a_y = \mathrm{diag}(y^{1/2}, y^{-1/2}) : y > 0\}$, $K = \mathrm{SO}_2(\mathbb{R})$ and $B = NA$. Then every $g \in \mathrm{SL}_2(\mathbb{R})$ factors uniquely as*

$$g = b(x, y) k, \quad x + iy = g \cdot i, \quad k \in K,$$

with $b(x, y) = u_x a_y$ as in (2.8). In particular $\mathrm{SL}_2(\mathbb{R}) = NAK$ and B acts simply transitively on $\mathbb{H}$.

Proof. Given g , set $x + iy := g \cdot i$ and $b := b(x, y)$. Then $b^{-1}g \cdot i = b^{-1} \cdot (x + iy) = i$ by (2.8), so $k := b^{-1}g$ fixes i and hence lies in K by Lemma 2.1; thus $g = bk$. If $b_1 k_1 = b_2 k_2$ are two such factorisations then $b_2^{-1} b_1 = k_2 k_1^{-1} \in B \cap K$. An element of B is upper triangular with positive diagonal; if it is also orthogonal, its columns are orthonormal, which forces it to be the identity. So $b_1 = b_2$ and $k_1 = k_2$. Simple transitivity of B on $\mathbb{H}$ is the same statement: $b \cdot i$ determines and is determined by b . $\square$

Lemma 2.9 (Haar measure in these coordinates).

- (i) In the coordinates (2.8), $dm_B = y^{-2} dx dy$ is a left Haar measure on B , and B is not unimodular: $\Delta_B(b(x_0, y_0)) = y_0^{-1}$.
- (ii) Under $b \mapsto b \cdot i$ the measure m_B corresponds to the hyperbolic area $y^{-2} dx dy$ on $\mathbb{H}$, which is $\mathrm{SL}_2(\mathbb{R})$ -invariant.
- (iii) With m_K the probability Haar measure on K , the functional

$$f \mapsto \int_B \int_K f(bk) dm_K(k) dm_B(b) \quad (f \in C_c(\mathrm{SL}_2(\mathbb{R})))$$

is a left Haar measure on $\mathrm{SL}_2(\mathbb{R})$. In short $m_{\mathrm{SL}_2(\mathbb{R})} \propto m_B \times m_K$ in the coordinates $g = bk$.

Proof. (i) A computation with (2.8) gives

$$b(x_0, y_0) b(x, y) = b(y_0 x + x_0, y_0 y), \quad b(x, y) b(x_0, y_0) = b(x + y x_0, y y_0).$$

Left translation by $b(x_0, y_0)$ is therefore $(x, y) \mapsto (y_0 x + x_0, y_0 y)$, with Jacobian y_0^2 , and since $(y_0 y)^{-2} \cdot y_0^2 = y^{-2}$ the measure $y^{-2} dx dy$ is left invariant. Right translation is $(x, y) \mapsto (x + y x_0, y y_0)$, with Jacobian y_0 , and $(y y_0)^{-2} y_0 = y_0^{-1} y^{-2}$, whence $\Delta_B(b(x_0, y_0)) = y_0^{-1}$.

(ii) The first assertion is $b(x, y) \cdot i = x + iy$. For the second, let $g \in \mathrm{SL}_2(\mathbb{R})$ and $z' = g \cdot z$. By (2.6) the differential of $z \mapsto g \cdot z$ is multiplication by $j(g, z)^{-2}$, so as a real map its Jacobian determinant is $|j(g, z)|^{-4}$; and $\mathrm{Im} z' = \mathrm{Im} z |j(g, z)|^{-2}$ by (2.3). Hence

$$\frac{dx' dy'}{(\mathrm{Im} z')^2} = \frac{|j(g, z)|^{-4} dx dy}{(\mathrm{Im} z)^2 |j(g, z)|^{-4}} = \frac{dx dy}{(\mathrm{Im} z)^2}.$$

(iii) Call the functional I . It is positive and non-zero, so by uniqueness of Haar measure it suffices to prove left invariance. Fix $g_0 \in \mathrm{SL}_2(\mathbb{R})$ and $f \in C_c$. For $b \in B$ write, using Lemma 2.8, $g_0 b = \beta(b) \kappa(b)$ with $\beta(b) \in B$, $\kappa(b) \in K$. Then

$$I(f(g_0 \cdot)) = \int_B \int_K f(\beta(b) \kappa(b) k) dm_K(k) dm_B(b) = \int_B \int_K f(\beta(b) k) dm_K(k) dm_B(b),$$

by invariance of m_K . It remains to see that $\beta: B \rightarrow B$ preserves m_B . By Lemma 2.8, $\beta(b)$ is the unique element of B with $\beta(b) \cdot i = g_0 b \cdot i$, so under the

identification of (ii) the map β becomes $z \mapsto g_0 \cdot z$ on $\mathbb{H}$, which preserves hyperbolic area. Hence $I(f(g_0 \cdot)) = I(f)$. $\square$

Proposition 2.10. $\mathrm{SL}_2(\mathbb{Z})$ is a lattice in $\mathrm{SL}_2(\mathbb{R})$.

Proof. Note that $\mathrm{SL}_2(\mathbb{Z}) \backslash \mathrm{SL}_2(\mathbb{R}) = \mathrm{PSL}_2(\mathbb{Z}) \backslash \mathrm{PSL}_2(\mathbb{R})$. Since $G = \mathrm{PSL}_2(\mathbb{R})$ is unimodular, by Lemma 2.6 and Proposition 1.17, it suffices to prove $m_G(F) < \infty$ for the set F of (2.10). By Lemma 2.8 we may write $g = b(x, y)k_\theta$ uniquely, and then $g \cdot i = b(x, y) \cdot i = x + iy$; so the condition $g \in F$ reads $x + iy \in E$, with θ unconstrained. In PSL_2 the angle θ ranges over $[0, \pi)$. Hence by Lemma 2.9(iii), up to the choice of normalisation,

$$m_G(F) = \iiint_E \int_0^\pi \frac{1}{y^2} d\theta dx dy \leq \int_{-1/2}^{1/2} \int_{\sqrt{3}/2}^\infty \int_0^\pi \frac{1}{y^2} d\theta dy dx = \frac{2\pi}{\sqrt{3}} < \infty,$$

where we used that E is contained in the strip $|x| \leq \frac{1}{2}$, $y \geq \frac{\sqrt{3}}{2}$. $\square$

Geometrically, X_2 is (apart from the two orbifold points coming from i and $\frac{1}{2} + \frac{\sqrt{3}}{2}i$, which have extra stabilisers) the unit tangent bundle of the modular surface $\mathrm{PSL}_2(\mathbb{Z}) \backslash \mathbb{H}$. That surface is obtained from Figure 5 by gluing the two vertical sides by $\begin{pmatrix} 1 & \pm 1 \\ 0 & 1 \end{pmatrix}$ and folding the circular side onto itself by $\begin{pmatrix} 1 & \\ 0 & -1 \end{pmatrix}$; it has one end, the *cusp*, corresponding to $y \rightarrow \infty$. In particular X_2 is not compact.

2.4. The geodesic flow — the subgroup A . Having shown in Proposition 2.10 that $\mathrm{SL}_2(\mathbb{Z})$ is a lattice in $\mathrm{SL}_2(\mathbb{R})$, so that the space $X_2 = \mathrm{SL}_2(\mathbb{Z}) \backslash \mathrm{SL}_2(\mathbb{R})$ carries an invariant probability measure m_{X_2} , we can at last do some dynamics. We begin with the diagonal subgroup, for two reasons: its action is the most geometric of all — it is the geodesic flow on the unit tangent bundle of the modular surface — and its orbits already display all the behaviours that the rest of the course is about.

So consider the diagonal subgroup

$$A = \left\{ a_t = \begin{pmatrix} e^{-t/2} & \\ & e^{t/2} \end{pmatrix} : t \in \mathbb{R} \right\},$$

also called a *Cartan subgroup*, and the associated flow on X_2 ,

$$(2.11) \quad a_t : X_2 \ni x \mapsto a_t \cdot x = x a_t^{-1} = x \begin{pmatrix} e^{t/2} & \\ & e^{-t/2} \end{pmatrix}.$$

Recall that the geodesic flow $(\varphi_t)_{t \in \mathbb{R}}$ on $T^1\mathbb{H}$ is defined geometrically: starting from $(z, v) \in T^1\mathbb{H}$, follow the unique geodesic through z tangent to v for hyperbolic distance $|t|$, forwards if $t > 0$ and backwards if $t < 0$, and let $\varphi_t(z, v)$ be the endpoint together with the tangent vector of the geodesic there.

Lemma 2.11. *Under the identification $\mathrm{PSL}_2(\mathbb{R}) \cong T^1\mathbb{H}$, $g \mapsto Dg(i, \uparrow)$, of Lemma 2.2, the geodesic flow becomes right multiplication by a_t^{-1} :*

$$\varphi_t(Dg(i, \uparrow)) = D(g a_t^{-1})(i, \uparrow) \quad (g \in \mathrm{PSL}_2(\mathbb{R}), t \in \mathbb{R}).$$

Consequently φ_t descends to $\Gamma \backslash \mathrm{PSL}_2(\mathbb{R})$ for every discrete Γ , and on X_2 it is the flow (2.11).

Proof. First take $g = e$. By Lemma 2.4 the geodesic through $(i, \uparrow)$ is $t \mapsto e^t i$, parametrised by arclength, so $\varphi_t(i, \uparrow) = (e^t i, \uparrow)$. On the other hand $a_t^{-1} = \begin{pmatrix} e^{t/2} & \\ & e^{-t/2} \end{pmatrix}$ acts on $\mathbb{H}$ by $z \mapsto e^t z$ and, by (2.6), on tangent vectors by $v \mapsto e^t v$. Since $\uparrow = i$ at the point i and $\uparrow = e^t i$ at the point $e^t i$,

$$D a_t^{-1}(i, \uparrow) = D a_t^{-1}(i, i) = (e^t i, e^t i) = (e^t i, \uparrow).$$

So the two prescriptions agree at $(i, \uparrow)$.

For general g we use that g acts by isometries (Lemma 2.1), so that Dg maps geodesics to geodesics preserving arclength and therefore commutes with the geodesic flow, $\varphi_t \circ Dg = Dg \circ \varphi_t$. Combining this with $Dg \circ Dh = D(gh)$ gives

$$\varphi_t(Dg(i, \uparrow)) = Dg(\varphi_t(i, \uparrow)) = Dg(D a_t^{-1}(i, \uparrow)) = D(g a_t^{-1})(i, \uparrow).$$

Right multiplication commutes with the left action of Γ , so the flow descends to $\Gamma \backslash \mathrm{PSL}_2(\mathbb{R})$, where by (2.11) it is $x \mapsto a_t \cdot x$. $\square$

So X_2 is the unit tangent bundle of the modular surface and A moves a unit vector along its own geodesic.

Next we shall see that A acts ergodically on X_2 , and in fact mixing: in the mean, every orbit is as equidistributed as it could be. Individual orbits, however, behave in wildly different ways, and the following list is instructive. The first two items will be analysed in §2.6 and §2.7.

- *Divergent trajectories*, for example $\mathrm{SL}_2(\mathbb{Z})A$, which corresponds to the vertical geodesic through $(i, \uparrow)$, i.e. to the lattices $a_t \mathbb{Z}^2$ marching into the cusp in both time directions (Figure 6, left).

- *Compact (equivalently: periodic) trajectories*, for example $\mathrm{SL}_2(\mathbb{Z})gA$, where $g \in \mathrm{SO}_2(\mathbb{R})$ diagonalises the symmetric matrix $\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$:

$$g^{-1} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} g = \begin{pmatrix} \frac{3-\sqrt{5}}{2} & \\ & \frac{3+\sqrt{5}}{2} \end{pmatrix} \in A.$$

Then $\mathrm{SL}_2(\mathbb{Z})ga_\ell = \mathrm{SL}_2(\mathbb{Z})g$ for $\ell = 2 \log \frac{3+\sqrt{5}}{2} = 4 \log \frac{1+\sqrt{5}}{2}$, so the orbit is the continuous image of a circle (Figure 6, right).

- *Dense trajectories*. These are much more numerous than the equidistributed ones, which by the pointwise ergodic theorem already account for m_{X_2} -almost every point.
- *Everything else*. There are A -orbits which are neither divergent nor compact nor dense. For instance, an orbit could be divergent backwards and bounded forwards, or bounded in one direction and equidistributed in the other. Moreover, the orbit closures of the geodesic flow are in general not embedded submanifolds — “nice” objects of integer dimension — but realise a dense set of Hausdorff dimensions in $[1, 3]$, a typical one of them being a fractal. The invariant measures are no better behaved: apart from m_{X_2} and the arclength measures on the compact orbits, the flow carries uncountably many mutually singular ergodic invariant probability measures and their entropies fill the whole interval $[0, 1]$. For the horocycle flow that we shall study later, by contrast, the situation is much more rigid and algebraic: the orbit closures are either compact orbits or all of X_2 , and the ergodic invariant measures are only the arclength measures on the periodic orbits and m_{X_2} .

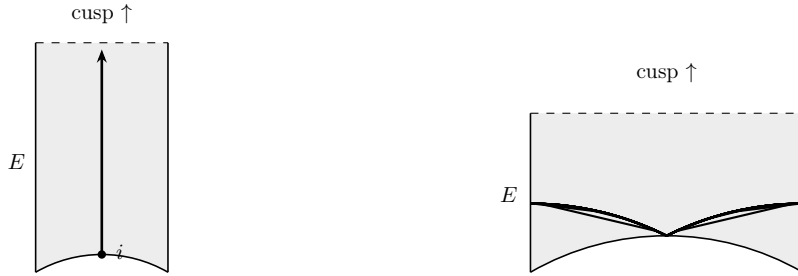


FIGURE 6. Two A -orbits, projected to the modular surface and drawn in the fundamental domain E . Left: the divergent orbit of $\mathrm{SL}_2(\mathbb{Z})$ itself, the vertical geodesic through i ; the backward half folds onto the forward half, since $S \cdot e^t i = e^{-t} i$, so the picture is the segment from i into the cusp traversed twice. Right (drawn on a larger scale, and only up to height 1.45): the compact orbit $\mathrm{SL}_2(\mathbb{Z})gA$, of length $4 \log \frac{1+\sqrt{5}}{2}$; in $\mathbb{H}$ it is the semicircle with endpoints the two roots of $z^2 + z - 1 = 0$, and the folding cuts it into the arcs shown. It is the shortest closed geodesic on the modular surface, and stays as far from the cusp as a closed geodesic can.

2.5. Ergodicity and mixing of the geodesic flow. We now prove that the geodesic flow is ergodic and mixing. The route is the following: translate the dynamical statements into statements about a unitary representation, and then prove a theorem about representations of $\mathrm{SL}_2(\mathbb{R})$. It is the first instance in this

course of a general principle that the dynamics of G on $\Gamma \backslash G$ is governed by the representation theory of G .

In this subsection, G is a locally compact, σ -compact group acting continuously on a locally compact Hausdorff probability space (X, μ) , and $H \leq G$ is a closed subgroup. We say the action is *measure-preserving* if $\mu(h \cdot B) = \mu(B)$ for every $h \in H$ and every Borel $B \subseteq X$.

Definition 2.12 (Ergodic, mixing). Suppose that H preserves μ .

- (1) The action is *ergodic* if every H -invariant Borel set has measure 0 or 1; here B is *H -invariant* if $\mu(h \cdot B \triangle B) = 0$ for every $h \in H$.
- (2) The action is *mixing* if for all Borel $B, C \subseteq X$

$$\mu(B \cap h \cdot C) \longrightarrow \mu(B) \mu(C) \quad \text{as } h \rightarrow \infty \text{ in } H,$$

where $h \rightarrow \infty$ means that h eventually leaves every compact subset of H .

Mixing is the assertion that the position of a point at time h becomes asymptotically independent of where it started; it implies ergodicity, since for an invariant B one may take $C = B$ and obtain $\mu(B) = \mu(B)^2$. For $H = A \cong \mathbb{R}$, “ $h \rightarrow \infty$ ” means $|t| \rightarrow \infty$.

Both properties are properties of a single unitary representation. We first recall the definition of a unitary operator.

Definition 2.13 (Unitary operator). Let $\mathcal{H}$ be a complex Hilbert space. A linear map $U: \mathcal{H} \rightarrow \mathcal{H}$ is *unitary* if it is bijective and preserves the inner product,

$$\langle Uv, Uw \rangle = \langle v, w \rangle \quad \text{for all } v, w \in \mathcal{H}.$$

Equivalently, U is bijective with $U^{-1} = U^*$; and equivalently again, U is a surjective linear isometry, $\|Uv\| = \|v\|$ for all v . The unitary operators form a group $\mathcal{U}(\mathcal{H})$ under composition.

Definition 2.14 (Unitary representation). Let $\mathcal{H}$ a complex Hilbert space, and write $\mathcal{U}(\mathcal{H})$ for the group of unitary operators on $\mathcal{H}$. A *unitary representation* of G on $\mathcal{H}$ is a group homomorphism $\pi: G \rightarrow \mathcal{U}(\mathcal{H})$ which is *strongly continuous*: for every $v \in \mathcal{H}$ the map

$$G \longrightarrow \mathcal{H}, \quad g \longmapsto \pi(g)v,$$

is continuous. A vector v is $\pi(G)$ -*invariant*, or *fixed*, if $\pi(g)v = v$ for all $g \in G$, and the functions $g \mapsto \langle \pi(g)v, w \rangle$ are the *matrix coefficients* of π .

Remark 2.15. For unitary π , strong continuity is equivalent to weak continuity, i.e. to the continuity of all matrix coefficients: this follows from

$$\|\pi(g)v - v\|^2 = 2\|v\|^2 - 2\operatorname{Re} \langle \pi(g)v, v \rangle$$

and the group law. We shall use both descriptions interchangeably.

The representation attached to a measure-preserving action is the following one.

Lemma 2.16 (Koopman representation). Suppose that H preserves μ . Then

$$\pi(h)f := f \circ h^{-1}, \quad \text{that is} \quad (\pi(h)f)(x) = f(h^{-1} \cdot x),$$

defines a unitary representation of H on $L^2(X, \mu)$.

Proof. Well defined. If $f = f'$ outside a null set N , then $f \circ h^{-1} = f' \circ h^{-1}$ outside $h \cdot N$, which is null because μ is H -invariant. So $\pi(h)$ is well defined on equivalence classes.

Homomorphism. For $h_1, h_2 \in H$,

$$\pi(h_1)\pi(h_2)f = (f \circ h_2^{-1}) \circ h_1^{-1} = f \circ (h_1 h_2)^{-1} = \pi(h_1 h_2)f,$$

which is exactly what the inverse in the definition is for.

Unitary. Substituting $y = h^{-1} \cdot x$ and using $h_*\mu = \mu$,

$$\|\pi(h)f\|_2^2 = \int_X |f(h^{-1} \cdot x)|^2 d\mu(x) = \int_X |f(y)|^2 d\mu(y) = \|f\|_2^2,$$

so $\pi(h)$ is a linear isometry; it is surjective because $\pi(h^{-1})$ is an inverse, and a surjective linear isometry of a Hilbert space preserves inner products by polarisation, hence is unitary.

Strong continuity. Since $\|\pi(h)\| = 1$ for every h , and since $C_c(X)$ is dense in $L^2(X, \mu)$, it suffices to prove continuity of $h \mapsto \pi(h)f$ for $f \in C_c(X)$: given $f \in L^2$ and $\varepsilon > 0$, choose $f' \in C_c(X)$ with $\|f - f'\|_2 < \varepsilon$ and estimate

$$\begin{aligned} \|\pi(h)f - \pi(h_0)f\|_2 &\leq \|\pi(h)(f - f')\|_2 + \|\pi(h)f' - \pi(h_0)f'\|_2 + \|\pi(h_0)(f' - f)\|_2 \\ &\leq 2\varepsilon + \|\pi(h)f' - \pi(h_0)f'\|_2. \end{aligned}$$

So let $f \in C_c(X)$ with support K , let $h_0 \in H$ and let V be a compact neighbourhood of h_0 . For $h \in V$ the function $\pi(h)f$ is supported in $V \cdot K$, which is compact. We claim that

$$(2.12) \quad \sup_{x \in X} |f(h^{-1} \cdot x) - f(h_0^{-1} \cdot x)| \longrightarrow 0 \quad (h \rightarrow h_0).$$

If not, there are $\varepsilon > 0$, $h_n \rightarrow h_0$ and $x_n \in X$ with $|f(h_n^{-1} \cdot x_n) - f(h_0^{-1} \cdot x_n)| \geq \varepsilon$. Then $h_n^{-1} \cdot x_n \in K$ or $h_0^{-1} \cdot x_n \in K$, so that $x_n \in V \cdot K$ for large n ; passing to a subsequence, $x_n \rightarrow x$ by compactness, and the joint continuity of the action together with the continuity of f makes both terms converge to $f(h_0^{-1} \cdot x)$, a contradiction. Finally, since μ is a probability measure, (2.12) gives

$$\|\pi(h)f - \pi(h_0)f\|_2 \leq \sup_{x \in X} |f(h^{-1} \cdot x) - f(h_0^{-1} \cdot x)| \longrightarrow 0. \quad \square$$

Write

$$L_0^2(X, \mu) = \left\{ f \in L^2(X, \mu) : \int_X f d\mu = 0 \right\},$$

the orthogonal complement of the constants; it is $\pi(H)$ -invariant.

Lemma 2.17 (Spectral criteria). *With the notation above:*

- (i) *the action of H is ergodic if and only if the only $\pi(H)$ -invariant vectors in $L^2(X, \mu)$ are the constants;*
- (ii) *the action of H is mixing if and only if*

$$\langle \pi(h)f, f' \rangle \longrightarrow 0 \quad (h \rightarrow \infty \text{ in } H)$$

for all $f, f' \in L_0^2(X, \mu)$; that is, if and only if the matrix coefficients of $\pi|_{L_0^2}$ vanish at infinity.

Proof. (i) If B is H -invariant then $\mathbf{1}_B$ is a $\pi(H)$ -invariant vector; if the only invariant vectors are constants then $\mathbf{1}_B$ is a.e. constant, so $\mu(B) \in \{0, 1\}$. Conversely let f be $\pi(H)$ -invariant and suppose the action is ergodic. For $c \in \mathbb{R}$ the set $\{f > c\}$ is H -invariant, hence of measure 0 or 1; letting $c_0 = \sup\{c : \mu(f > c) = 1\}$ gives $f = c_0$ almost everywhere.

(ii) Since $\langle \pi(h)\mathbf{1}_C, \mathbf{1}_B \rangle = \mu(B \cap h \cdot C)$, the condition in Definition 2.12(ii) says exactly that $\langle \pi(h)f, f' \rangle \rightarrow \langle f, 1 \rangle \langle f', 1 \rangle$ for f, f' indicators, i.e. that $\langle \pi(h)f_0, f'_0 \rangle \rightarrow 0$ for their projections f_0, f'_0 to L_0^2 . Simple functions are dense in L^2 and the operators $\pi(h)$ have norm 1, so the general case follows by a three- ε argument: if $\|f_0 - p\|, \|f'_0 - p'\| < \varepsilon$ with p, p' simple, then $|\langle \pi(h)f_0, f'_0 \rangle| \leq |\langle \pi(h)p, p' \rangle| + 2\varepsilon \cdot C$ uniformly in h . $\square$

So everything reduces to the following question about representations: *when do the matrix coefficients of a unitary representation of $\mathrm{SL}_2(\mathbb{R})$ vanish at infinity?* The answer is as clean as one could wish: always, unless there is an invariant vector.

Two facts about $\mathrm{SL}_2(\mathbb{R})$ and about unitary representations. We use three standard facts, the first two about the group and the third from functional analysis.

Lemma 2.18 (Cartan decomposition). $\mathrm{SL}_2(\mathbb{R}) = KA^+K$ with $K = \mathrm{SO}_2(\mathbb{R})$ and $A^+ = \{a_t : t \geq 0\}$. Moreover a sequence $g_n = k_n a_{t_n} k'_n$ tends to infinity in $\mathrm{SL}_2(\mathbb{R})$ if and only if $t_n \rightarrow \infty$.

Proof. This is the singular value decomposition: for $g \in \mathrm{SL}_2(\mathbb{R})$ the matrix $g^T g$ is symmetric positive definite, so $g^T g = k d^2 k^{-1}$ with $k \in \mathrm{SO}_2(\mathbb{R})$ and d diagonal positive with $\det d = 1$; then $k' := d^{-1} k^{-1} g$ satisfies $k'^T k' = I$, so $g = k d k'$ with $k' \in \mathrm{O}_2(\mathbb{R})$, and adjusting signs we may take $k' \in \mathrm{SO}_2(\mathbb{R})$ and $d = a_t$ with $t \geq 0$. Since K is compact, $\|g_n\| \rightarrow \infty$ if and only if $\|a_{t_n}\| = e^{t_n/2} \rightarrow \infty$. $\square$

Lemma 2.19. $\mathrm{SL}_2(\mathbb{R})$ is generated by the two unipotent subgroups

$$N^+ = \left\{ u_s = \begin{pmatrix} 1 & s \\ & 1 \end{pmatrix} \right\}, \quad N^- = \left\{ u_s^- = \begin{pmatrix} 1 & \\ s & 1 \end{pmatrix} \right\},$$

and it is perfect, i.e. equal to its own commutator subgroup. In particular $\mathrm{SL}_2(\mathbb{R})$ has no non-trivial continuous homomorphism into an abelian group.

Proof. Generation is Gaussian elimination: every matrix of determinant 1 is a product of elementary matrices. For perfectness, note that

$$\begin{pmatrix} t & \\ & t^{-1} \end{pmatrix} u_s \begin{pmatrix} t^{-1} & \\ & t \end{pmatrix} u_{-s} = u_{(t^2-1)s},$$

so every u_r is a commutator; the same holds in N^- , and these generate. A homomorphism into an abelian group kills commutators. $\square$

Finally we shall use that the closed unit ball of a Hilbert space is weakly compact and, since $L^2(X, \mu)$ is separable in our applications, weakly sequentially compact: every bounded sequence has a weakly convergent subsequence.

The contraction lemma. Everything now rests on one lemma, and on the elementary observation that conjugation by the geodesic direction contracts the two unipotent directions, one in forward and one in backward time:

$$(2.13) \quad a_t u_s a_t^{-1} = u_{e^{-t}s}, \quad a_t u_s^- a_t^{-1} = u_{e^t s}^-.$$

Thus $a_{-t} u_s^- a_t \rightarrow e$ and $a_t u_s a_{-t} \rightarrow e$ as $t \rightarrow +\infty$.

Lemma 2.20 (Contraction lemma). *Let π be a strongly continuous unitary representation of a topological group G on a Hilbert space $\mathcal{H}$, let (g_n) be a sequence in G , and let $v \in \mathcal{H}$ be such that $\pi(g_n)v$ converges weakly to some $z \in \mathcal{H}$. Then z is fixed by every $g \in G$ with*

$$g_n^{-1} g g_n \longrightarrow e.$$

Proof. Let $w \in \mathcal{H}$. Using $\pi(g)\pi(g_n) = \pi(g_n)\pi(g_n^{-1}gg_n)$ and the fact that $\pi(g_n)$ is unitary,

$$|\langle \pi(g)z - z, w \rangle| = \lim_{n \rightarrow \infty} |\langle \pi(g_n)(\pi(g_n^{-1}gg_n)v - v), w \rangle| = \lim_{n \rightarrow \infty} |\langle \pi(g_n^{-1}gg_n)v - v, \pi(g_n)^{-1}w \rangle|,$$

which by Cauchy–Schwarz is at most $\lim_n \|\pi(g_n^{-1}gg_n)v - v\| \|w\| = 0$, by strong continuity. As w was arbitrary, $\pi(g)z = z$. $\square$

The lemma is worth reading twice, because it is the mechanism behind everything in this subsection: *a weak limit of translates is invariant in every direction which the translations contract*. Its first consequence is classical and costs one line.

Corollary 2.21 (Mautner phenomenon). *Let π be a strongly continuous unitary representation of $\mathrm{SL}_2(\mathbb{R})$. Every A -fixed vector is G -fixed.*

Proof. Let v be fixed by A . Apply Lemma 2.20 to the sequence $g_n = a_n$, for which $\pi(g_n)v = v$ converges weakly to v : by (2.13) we have $a_{-n}u_s^-a_n \rightarrow e$, so v is N^- -fixed. Applying it instead to $g_n = a_{-n}$ gives that v is N^+ -fixed. By Lemma 2.19 the two unipotent subgroups generate G , so v is G -fixed. $\square$

Ergodicity of the geodesic flow follows at once, and it is worth noting how little it costs: no representation theory beyond Lemma 2.20, and the fact that G acts on X_2 transitively.

Corollary 2.22 (Ergodicity of the geodesic flow). *A acts ergodically on (X_2, m_{X_2}) .*

Proof. Let $f \in L^2(X_2)$ be A -invariant, that is, $\pi(a_t)f = f$ for every t , where π is the Koopman representation of $\mathrm{SL}_2(\mathbb{R})$. By Corollary 2.21, f is $\pi(G)$ -invariant, i.e. $f(xg) = f(x)$ for almost every x and every $g \in G$. Since G acts transitively on $X_2 = \Gamma \backslash G$ — the orbit of Γe is all of X_2 — such an f is almost everywhere constant. By Lemma 2.17(i) the action is ergodic. $\square$

The Howe–Moore theorem. Ergodicity came cheap. Mixing is a genuinely stronger statement, and the theorem behind it is the following one; remarkably, the proof needs nothing beyond the contraction lemma, applied twice.

Theorem 2.23 (Howe–Moore). *Let π be a strongly continuous unitary representation of $G = \mathrm{SL}_2(\mathbb{R})$ on a separable Hilbert space $\mathcal{H}$ with no non-zero G -invariant vector. Then all matrix coefficients vanish at infinity:*

$$\langle \pi(g)v, w \rangle \longrightarrow 0 \quad \text{as } g \rightarrow \infty \text{ in } G,$$

for all $v, w \in \mathcal{H}$.

Proof. Suppose the conclusion fails. Then there are $v, w \in \mathcal{H}$, a constant $c \neq 0$ and a sequence $g_n \rightarrow \infty$ in G with $\langle \pi(g_n)v, w \rangle \rightarrow c$.

Step 1: reduction to the geodesic direction. Write $g_n = k_n a_{t_n} k'_n$ by Lemma 2.18, so that $t_n \rightarrow \infty$; passing to a subsequence we may assume $k_n \rightarrow k$ and $k'_n \rightarrow k'$ in the compact group K . By strong continuity $\pi(k'_n)v \rightarrow \pi(k')v$ and $\pi(k_n)^{-1}w \rightarrow \pi(k)^{-1}w$ in norm, so setting $v_0 := \pi(k')v$ and $w_0 := \pi(k)^{-1}w$ we obtain

$$(2.14) \quad \langle \pi(a_{t_n})v_0, w_0 \rangle \longrightarrow c \neq 0.$$

Step 2: a weak limit, fixed by N^- . The sequence $\pi(a_{t_n})v_0$ is bounded, so after passing to a further subsequence it converges weakly to some $z \in \mathcal{H}$. By (2.14) we have $\langle z, w_0 \rangle = c \neq 0$, so

$$(2.15) \quad z \neq 0.$$

Since $a_{-t}u_s^-a_t = u_{e^{-t}s}^- \rightarrow e$ as $t \rightarrow +\infty$, the contraction lemma applied to the sequence (a_{t_n}) shows that z is fixed by N^- .

Step 3: the subspace of N^- -fixed vectors is A -invariant. Write $\mathcal{H}^{N^-}$ for that subspace; it is a closed, hence weakly closed, subspace. If $y \in \mathcal{H}^{N^-}$ and $t \in \mathbb{R}$, then by (2.13)

$$\pi(u_s^-)\pi(a_t)y = \pi(a_t)\pi(a_t^{-1}u_s^-a_t)y = \pi(a_t)\pi(u_{e^{-t}s}^-)y = \pi(a_t)y,$$

so $\pi(a_t)y \in \mathcal{H}^{N^-}$ as well.

Step 4: the limit is annihilated by running the flow backwards. Apply the contraction lemma a second time, now to the vector z and the sequence (a_{-t_n}) : passing to a further subsequence, assume $\pi(a_{-t_n})z \rightarrow z'$. Since $a_t u_s a_{-t} = u_{e^{-t}s}^- \rightarrow e$ as $t \rightarrow +\infty$, the limit z' is fixed by N^+ . By Step 3 each $\pi(a_{-t_n})z$ lies in $\mathcal{H}^{N^-}$, which is weakly closed, so z' is fixed by N^- as well. By Lemma 2.19 it is therefore G -fixed, and by hypothesis

$$z' = 0.$$

Step 5: the contradiction. Since $\pi(a_{-t_n})$ is unitary with adjoint $\pi(a_{t_n})$, we may compute the same limit in two ways:

$$0 = \lim_{n \rightarrow \infty} \langle \pi(a_{-t_n})z, v_0 \rangle = \lim_{n \rightarrow \infty} \langle z, \pi(a_{t_n})v_0 \rangle = \langle z, z \rangle = \|z\|^2,$$

the first equality because $\pi(a_{-t_n})z \rightarrow z' = 0$, the third because $\pi(a_{t_n})v_0 \rightarrow z$. Hence $z = 0$, contradicting (2.15). $\square$

Moore's ergodicity theorem and the consequences.

Corollary 2.24 (Moore's ergodicity theorem). *Let π be as in Theorem 2.23 and let $H \leq G$ be a closed unbounded subgroup. Then $\pi|_H$ has no non-zero invariant vector, and its matrix coefficients vanish at infinity along H .*

Proof. If v were an H -invariant unit vector, then $\langle \pi(h)v, v \rangle = 1$ for every $h \in H$, contradicting Theorem 2.23 along a sequence $h_n \rightarrow \infty$ in H , which exists because H is unbounded and closed. The second assertion is Theorem 2.23 restricted to H . $\square$

Corollary 2.25. *The action of $\mathrm{SL}_2(\mathbb{R})$ on (X_2, m_{X_2}) is mixing, and so is the action of every closed unbounded subgroup $H \leq \mathrm{SL}_2(\mathbb{R})$. In particular:*

- (i) *the geodesic flow (a_t) is mixing on X_2 , hence ergodic;*
- (ii) *the horocycle flow (u_s) is mixing on X_2 , hence ergodic;*
- (iii) *more generally, every closed unbounded $H \leq \mathrm{SL}_2(\mathbb{R})$ acts ergodically on X_2 ; the only closed subgroups which do not are the bounded ones, that is, the subgroups of conjugates of $K = \mathrm{SO}_2(\mathbb{R})$.*

Proof. Apply Theorem 2.23 to the Koopman representation of $\mathrm{SL}_2(\mathbb{R})$ on $L_0^2(X_2, m_{X_2})$. There is no non-zero G -invariant vector there: a G -invariant $f \in L^2$ is constant, because G acts transitively on X_2 , and the constants in L_0^2 are 0. So all matrix coefficients on L_0^2 vanish at infinity, which by Lemma 2.17(ii) is exactly mixing. Restricting to an unbounded closed H gives the same conclusion along H , by Corollary 2.24. Finally a closed bounded subgroup of $\mathrm{SL}_2(\mathbb{R})$ has compact closure and is therefore contained in a conjugate of the maximal compact subgroup K , whose orbits on X_2 are compact and certainly not dense. $\square$

We notice that the only property of $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ that entered is that X_2 carries an invariant probability measure; the same proof gives ergodicity and mixing of the geodesic and horocycle flows on $\Gamma \backslash \mathrm{SL}_2(\mathbb{R})$ for *every* lattice Γ , cocompact or not. This is the strength and the weakness of the spectral method: it says something about almost every orbit and nothing whatever about any particular one. The rest of this chapter is concerned with the orbits it cannot see — the divergent ones in §2.6, the compact ones in §2.7 — and with the fact, established later, that for A there is no classification of the remainder.

2.6. Divergent A -orbits and rationality. The following elementary lemma will be used twice; it says that for one-parameter flows on G/Γ , “compact orbit” and “periodic orbit” mean the same thing.

Lemma 2.26. *Let G be a locally compact metrisable group, $\Gamma < G$ discrete, and let $(h_t)_{t \in \mathbb{R}}$ be a continuous one-parameter subgroup of G with $h_t \neq e$ for $t \neq 0$. Let $x = \Gamma g \in G/\Gamma$ and suppose that the orbit $Hx = \{\Gamma gh_t : t \in \mathbb{R}\}$ is compact. Then*

$$\Sigma_x := \{t \in \mathbb{R} : xh_t = x\} = \{t \in \mathbb{R} : gh_tg^{-1} \in \Gamma\}$$

equals $\ell\mathbb{Z}$ for some $\ell > 0$, and $t \mapsto xh_t$ induces a homeomorphism $\mathbb{R}/\ell\mathbb{Z} \rightarrow xH$.

Proof. Σ_x is a closed subgroup of $\mathbb{R}$. It is discrete: otherwise there are $t_n \rightarrow 0$ with $t_n \neq 0$ and $\gamma_n := gh_{t_n}g^{-1} \in \Gamma \setminus \{e\}$, and $\gamma_n \rightarrow e$, contradicting the discreteness of Γ . Hence $\Sigma_x = \{0\}$ or $\Sigma_x = \ell\mathbb{Z}$ with $\ell > 0$. In the latter case the orbit map induces a continuous bijection $\mathbb{R}/\ell\mathbb{Z} \rightarrow xH$ from a compact space to a Hausdorff space, hence a homeomorphism.

Assume then $\Sigma_x = \{0\}$, so that $\psi: \mathbb{R} \rightarrow xH$, $t \mapsto xh_t$, is a continuous bijection; we derive a contradiction. The space xH is compact metrisable, hence a Baire space, and $xH = \bigcup_{n \geq 1} \psi([-n, n])$ with each $\psi([-n, n])$ closed. So some $\psi([-n, n])$ contains a non-empty open subset V of xH . Since $\psi|_{[-n, n]}$ is a continuous bijection from a compact space onto its image, it is a homeomorphism onto $\psi([-n, n])$; therefore, for $t_0 \in \psi^{-1}(V)$ in the interior of $[-n, n]$ and $\delta > 0$ small, the set $W := \psi((t_0 - \delta, t_0 + \delta)) \cap V$ is open in $\psi([-n, n])$ and contained in V , hence open in xH (as V is open in xH). Thus $\psi(t_0)$ has a neighbourhood in xH contained in $\psi((t_0 - \delta, t_0 + \delta))$. Translating by the homeomorphisms $y \mapsto yh_s$ of xH , the same holds at every point, so ψ is an open map and hence a homeomorphism. But then xH is homeomorphic to $\mathbb{R}$, which is not compact — a contradiction. $\square$

Proposition 2.27. *Let $g \in \mathrm{SL}_2(\mathbb{R})$. The trajectory $\mathrm{SL}_2(\mathbb{Z})gA$ is divergent, i.e. the orbit map $A \ni a \mapsto \mathrm{SL}_2(\mathbb{Z})ga$ is proper, if and only if*

$$ga \in \mathrm{SL}_2(\mathbb{Q}) \quad \text{for some } a \in A.$$

Proof. We work with the lattice $\Lambda := \Lambda_{\Gamma g} = g^{-1}\mathbb{Z}^2$ of Section 1.2, whose A -orbit is $\{a_t\Lambda : t \in \mathbb{R}\}$, and we write $v = (v_1, v_2)$, so that

$$(2.16) \quad \|a_tv\|^2 = e^{-t}v_1^2 + e^tv_2^2 =: f_v(t).$$

By Mahler’s criterion (Section 1.2(c)), properness of the orbit map is equivalent to $\lambda_1(a_t\Lambda) \rightarrow 0$ as $|t| \rightarrow \infty$.

Sufficiency. If $h := (ga)^{-1} \in \mathrm{SL}_2(\mathbb{Q})$ then, choosing $M \in \mathbb{N}$ with $Mh^{-1} \in \mathrm{Mat}_2(\mathbb{Z})$, we get $M\mathbb{Z}^2 \subseteq h\mathbb{Z}^2 = a^{-1}\Lambda$. Hence $a^{-1}\Lambda$ contains Me_1 and Me_2 , so by (2.16)

$$\lambda_1(a_t a^{-1}\Lambda) \leq M e^{-|t|/2} \xrightarrow{|t| \rightarrow \infty} 0,$$

and the orbit of Λ is a time-translate of the orbit of $a^{-1}\Lambda$, so it is divergent as well.

Necessity. Assume the orbit is divergent and fix $\varepsilon \in (0, 1)$ and $T > 0$ with $\lambda_1(a_t\Lambda) < \varepsilon$ for $|t| \geq T$. For $t \in \mathbb{R}$ put $V_t = \{v \in \Lambda \setminus \{0\} : \|a_tv\| < \varepsilon\}$; thus $V_t \neq \emptyset$ for $|t| \geq T$. If $v, w \in V_t$, then

$$|\det(v, w)| = |\det(a_tv, a_tw)| \leq \|a_tv\| \|a_tw\| < \varepsilon^2 < 1;$$

since $\det(v, w) \in \mathrm{covol}(\Lambda)\mathbb{Z} = \mathbb{Z}$ for $v, w \in \Lambda$, we get $\det(v, w) = 0$, i.e. all elements of V_t lie on one line. Consequently, if for a primitive $v \in \Lambda$ we set

$$J_v = \{t \in \mathbb{R} : f_v(t) < \varepsilon^2\},$$

then $J_v \cap J_w = \emptyset$ whenever v, w are primitive and $w \neq \pm v$, while $J_v = J_{-v}$. Moreover each J_v is an open interval, since f_v is convex, and J_v is unbounded above if and only if $v_2 = 0$ (and unbounded below if and only if $v_1 = 0$). Finally, the J_v cover $[T, \infty)$: if $t \geq T$ and $v \in V_t$, then the primitive vector v' with $v \in \mathbb{N}v'$ satisfies $\|a_t v'\| \leq \|a_t v\| < \varepsilon$.

Now $[T, \infty)$ is connected and covered by the pairwise disjoint open sets J_v ; hence it is contained in a single one of them. That J_v is unbounded above, so Λ contains a non-zero vector $v = (v_1, 0)$, and we may take v primitive and $v_1 > 0$. Running the same argument on $(-\infty, -T]$ produces a primitive $w = (0, w_2) \in \Lambda$.

Then $\mathbb{Z}v \oplus \mathbb{Z}w \subseteq \Lambda$ has index $N := |\det(v, w)| = |v_1 w_2| \in \mathbb{N}$, whence $N\Lambda \subseteq \mathbb{Z}v \oplus \mathbb{Z}w$. Choose $s = 2 \log v_1$, so that $a_s v = e_1$ and $a_s w = (0, v_1 w_2) = \pm N e_2$. Then

$$\mathbb{Z}e_1 \oplus N\mathbb{Z}e_2 \subseteq a_s \Lambda \subseteq \frac{1}{N}(\mathbb{Z}e_1 \oplus N\mathbb{Z}e_2) \subseteq \frac{1}{N}\mathbb{Z}^2,$$

so $a_s \Lambda$ has a rational basis, i.e. $a_s \Lambda = h\mathbb{Z}^2$ with $h \in \text{Mat}_2(\mathbb{Q}) \cap \text{SL}_2(\mathbb{R}) = \text{SL}_2(\mathbb{Q})$. As $a_s \Lambda = (ga_s^{-1})^{-1}\mathbb{Z}^2$, this says $ga_s^{-1} \in \text{SL}_2(\mathbb{Q})$, as required. $\square$

It is worth isolating what made the argument work: in $\mathbb{R}^2$ two independent lattice vectors cannot both be short, so “the shortest vector” is unique up to sign, and a connectedness argument forces it to be *the same* vector for all large t . The analogous statement in $\mathbb{R}^d$, $d \geq 3$, is false — there exist divergent A -orbits that are not associated with rational data — and this is the phenomenon of *singular* vectors.

2.7. Compact A -orbits and real quadratic fields. Let $\Gamma = \text{SL}_2(\mathbb{Z})$.

Proposition 2.28 (Length of compact A -orbits). *Let $x = \Gamma g \in X_2$ and suppose the orbit xA is compact. Then there is a real quadratic number field K and a unit $\xi \in \mathcal{O}^\times$ of an order $\mathcal{O} \subseteq K$ such that the length of the orbit is $\log |\xi|$.*

Proof. By Lemma 2.26 there is $\ell > 0$ with $\Sigma_x = \ell\mathbb{Z}$; the orbit is the image of the circle $\mathbb{R}/\ell\mathbb{Z}$, and ℓ is its length with respect to the parametrisation (2.11). Set

$$\gamma := g a_\ell g^{-1} \in \Gamma = \text{SL}_2(\mathbb{Z}).$$

Then γ has eigenvalues $e^{\pm \ell/2} \notin \{\pm 1\}$, so $n := \text{tr } \gamma = 2 \cosh(\ell/2)$ is an integer with $n \geq 3$, and the characteristic polynomial of γ is

$$\chi(\lambda) = \lambda^2 - n\lambda + 1.$$

Its roots are $\varepsilon^{\pm 1}$ with $\varepsilon = \frac{n + \sqrt{n^2 - 4}}{2} = e^{\ell/2}$. Since $n \geq 3$, $n^2 - 4$ is positive and is not a square (if $n^2 - 4 = m^2$ then $(n - m)(n + m) = 4$, forcing $n = 2$), so

$$K := \mathbb{Q}(\sqrt{n^2 - 4})$$

is a real quadratic field, ε is an algebraic integer of norm 1, and hence a unit of the order $\mathcal{O} := \mathbb{Z}[\varepsilon] \subseteq K$. With our normalisation (2.11) the length of the orbit is

$$\ell = 2 \log \varepsilon = \log \xi, \quad \xi := \varepsilon^2 \in \mathcal{O}^\times,$$

which is the asserted statement. $\square$

We recall that an element $\gamma \in \text{SL}_2(\mathbb{Z})$ is called *hyperbolic* if $|\text{tr } \gamma| > 2$, and *primitive* if it is not a proper power of an element of Γ .

Proposition 2.29 (Compact A -orbits and primitive hyperbolic elements). *There is a bijective correspondence between compact A -orbits in X_2 and conjugacy classes in $\Gamma = \text{SL}_2(\mathbb{Z})$ of primitive hyperbolic elements $\gamma \in \Gamma$ with $\text{tr } \gamma > 2$.*

Proof. Suppose first that $x = \Gamma g \in X_2$ has compact A -orbit. The proof of Proposition 2.28 shows that the element $\gamma = ga_\ell g^{-1} \in \Gamma$ associated to the compact orbit xA is hyperbolic. We first check that the conjugacy class of γ depends only on the A -orbit. Replacing g by another representative δg , with $\delta \in \Gamma$, replaces γ by

$$(\delta g)a_\ell(\delta g)^{-1} = \delta\gamma\delta^{-1}.$$

On the other hand, replacing g by ga_s , that is, choosing another point on the same A -orbit, does not change γ , since A is abelian:

$$(ga_s)a_\ell(ga_s)^{-1} = ga_\ell g^{-1}.$$

Hence the compact A -orbit determines a well-defined conjugacy class in Γ . We next show that γ is primitive. Suppose, to the contrary, that $\gamma = \eta^k$ for some $\eta \in \Gamma$ and some integer $k \geq 2$. Since η commutes with γ , it preserves the two distinct eigenspaces of γ . Therefore, in the eigenbasis given by g ,

$$g^{-1}\eta g = \begin{pmatrix} r & 0 \\ 0 & r^{-1} \end{pmatrix}$$

for some $r \in \mathbb{R}^\times$. From $(g^{-1}\eta g)^k = g^{-1}\gamma g = a_\ell$ it follows that

$$g^{-1}\eta g = \pm a_{\ell/k}.$$

(The minus sign can occur only when k is even.) Since $-I \in \Gamma$, it follows in either case that $ga_{\ell/k}g^{-1} \in \Gamma$. Consequently $\Gamma ga_{\ell/k} = \Gamma g$, contradicting the minimality of ℓ . Thus γ is primitive.

Conversely, let $\gamma \in \Gamma$ be primitive and hyperbolic with

$$\text{tr } \gamma > 2.$$

Its characteristic polynomial is

$$p_\gamma(\lambda) = \lambda^2 - (\text{tr } \gamma)\lambda + 1.$$

Since $(\text{tr } \gamma)^2 - 4 > 0$, the polynomial has two distinct real roots. Thus γ has two linearly independent real eigenvectors and is diagonalizable over $\mathbb{R}$. Moreover, since

$$\det \gamma = 1 \quad \text{and} \quad \text{tr } \gamma > 2,$$

its eigenvalues are positive and reciprocal. Hence they have the form $e^{\pm \ell/2}$ for a unique $\ell > 0$. Choosing an oriented eigenbasis and rescaling one eigenvector if necessary, we may find $g \in \text{SL}_2(\mathbb{R})$ such that

$$g^{-1}\gamma g = a_\ell.$$

Therefore $\Gamma ga_\ell = \Gamma \gamma g = \Gamma g$, and hence the orbit ΓgA is periodic. Since it is the continuous image of $\mathbb{R}/\ell\mathbb{Z}$, it is compact.

We claim that ℓ is its least positive period. Let $s > 0$ denote the least positive period. Since ℓ is a period, there is an integer $k \geq 1$ such that $\ell = ks$. Setting $\eta := ga_s g^{-1} \in \Gamma$, we obtain

$$\gamma = ga_\ell g^{-1} = ga_{ks} g^{-1} = \eta^k.$$

By primitivity of γ , we must have $k = 1$, hence $s = \ell$.

It remains to verify injectivity. Suppose two compact A -orbits, represented by ΓgA and $\Gamma g'A$, determine the same conjugacy class. After replacing g' by $\delta g'$ for a suitable $\delta \in \Gamma$, we may assume that

$$ga_\ell g^{-1} = g'a_{\ell'} g'^{-1},$$

where $\ell, \ell' > 0$ are the respective least positive periods. Comparison of eigenvalues gives $\ell = \ell'$. It follows that

$$g'^{-1}g a_\ell = a_\ell g'^{-1}g,$$

so $g'^{-1}g$ belongs to the centralizer of a_ℓ in $\mathrm{SL}_2(\mathbb{R})$. Since a_ℓ has two distinct eigenvalues, any matrix commuting with it must preserve its two eigenspaces and hence must be diagonal. Therefore

$$C_{\mathrm{SL}_2(\mathbb{R})}(a_\ell) = \left\{ \begin{pmatrix} r & 0 \\ 0 & r^{-1} \end{pmatrix} : r \in \mathbb{R}^\times \right\} = \pm A.$$

Thus

$$g'^{-1}g \in \pm A.$$

Since $-I \in \Gamma$, multiplication by the sign does not change the point of X_2 , and consequently

$$\Gamma g' A = \Gamma g A.$$

This establishes the desired correspondence. $\square$

For instance, for $\gamma = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ one has $n = 3$, $K = \mathbb{Q}(\sqrt{5})$ and $\varepsilon = \frac{3+\sqrt{5}}{2} = \varphi^2$ with $\varphi = \frac{1+\sqrt{5}}{2}$ the golden ratio; this is the orbit mentioned above.

Corollary 2.30 (Countability). *There are only countably many compact A -orbits.*

We now show that every compact orbit is built out of arithmetic data. It is most transparent in the language of lattices, so we use the dictionary of Section 1.2 throughout: $x \in X_2$ corresponds to the unimodular lattice $\Lambda_x \subseteq \mathbb{R}^2$, and $a \cdot x$ corresponds to $a\Lambda_x$.

Notation. Let K be a real quadratic field. We fix an embedding $K \subseteq \mathbb{R}$ and write $\alpha \mapsto \bar{\alpha}$ for the non-trivial element of $\mathrm{Gal}(K/\mathbb{Q})$, so that the two embeddings $K \rightarrow \mathbb{R}$ are $\alpha \mapsto \bar{\alpha}$ and $\alpha \mapsto \alpha$, and $N(\alpha) = \alpha\bar{\alpha}$ is the norm. Define

$$\iota: K \longrightarrow \mathbb{R}^2, \quad \iota(\alpha) = (\bar{\alpha}, \alpha), \quad m_\alpha := \begin{pmatrix} \bar{\alpha} & \\ & \alpha \end{pmatrix} \in \mathrm{Mat}_2(\mathbb{R}).$$

Both embeddings are ring homomorphisms, so

$$(2.17) \quad \iota(\alpha\beta) = m_\alpha \iota(\beta), \quad m_{\alpha\beta} = m_\alpha m_\beta, \quad \det m_\alpha = N(\alpha) :$$

under ι , multiplication by α on K becomes the diagonal matrix m_α . Call $I \subseteq K$ a *lattice in K* if it is a finitely generated $\mathbb{Z}$ -submodule containing a $\mathbb{Q}$ -basis of K ; equivalently $I = \mathbb{Z}\mu_1 \oplus \mathbb{Z}\mu_2$ for some $\mathbb{Q}$ -basis μ_1, μ_2 of K . Since ι extends to an isomorphism $K \otimes_{\mathbb{Q}} \mathbb{R} \rightarrow \mathbb{R}^2$, the image $\iota(I)$ is then a lattice in $\mathbb{R}^2$. Its *ring of multipliers*

$$\mathcal{O}_I := \{\alpha \in K : \alpha I \subseteq I\}$$

is an order of K , i.e. a subring which is also a lattice in K . Indeed it is a subring containing $\mathbb{Z}$; it is contained in the lattice $\mu^{-1}I$ for any $\mu \in I \setminus \{0\}$, hence is a finitely generated $\mathbb{Z}$ -module; and it spans K (In fact, let $\alpha \in K$. Since μ_1, μ_2 form a $\mathbb{Q}$ -basis of K , we have $\alpha\mu_1 = q_{11}\mu_1 + q_{12}\mu_2$ and $\alpha\mu_2 = q_{21}\mu_1 + q_{22}\mu_2$ with $q_{ij} \in \mathbb{Q}$. Choose $k \in \mathbb{Z}_{>0}$ to clear the denominators: $k\alpha\mu_1, k\alpha\mu_2 \in I$, hence $k\alpha \in \mathcal{O}_I$). The group of units is $\mathcal{O}_I^\times = \{\delta \in \mathcal{O}_I : \delta^{-1} \in \mathcal{O}_I\}$ (note that $\delta \in \mathcal{O}_I$ is a unit if and only if it preserves I bijectively: $\delta I = I$). Finally set

$$\mathcal{O}_I^{\times,+} := \{\delta \in \mathcal{O}_I^\times : N(\delta) = 1, \delta > 0\};$$

note that $N(\delta) = 1$ and $\delta > 0$ force $\bar{\delta} = \delta^{-1} > 0$, so these are exactly the totally positive units (meaning units δ such that $\delta > 0$ and $\bar{\delta} > 0$) of norm one. By

Dirichlet's unit theorem $\mathcal{O}_I^\times = \{\pm 1\} \times \varepsilon_0^\mathbb{Z}$ for a *fundamental unit* $\varepsilon_0 > 1$, and $\mathcal{O}_I^{\times,+}$ is an infinite cyclic subgroup; we write $\varepsilon_I > 1$ for its generator. Explicitly,

$$(2.18) \quad \varepsilon_I = \begin{cases} \varepsilon_0, & \text{if } N(\varepsilon_0) = 1, \\ \varepsilon_0^2, & \text{if } N(\varepsilon_0) = -1, \end{cases}$$

since $\pm \varepsilon_0^k$ is totally positive of norm one precisely when the sign is $+$ and $N(\varepsilon_0)^k = 1$.

Finally, let $x_I \in X_2$ denote the point corresponding under Section 1.2(a) to the unimodular lattice $\text{covol}(\iota(I))^{-1/2} \iota(I)$.

Proposition 2.31 (Arithmetic description of compact A -orbits).

- (i) *Let K be a real quadratic field and $I \subseteq K$ any lattice in K . Then the orbit $x_I A$ is compact, and*

$$\Sigma_{x_I} = (2 \log \varepsilon_I) \mathbb{Z}.$$

In particular, the length of $x_I A$ is $2 \log \varepsilon_I$, and hence depends only on the order $\mathcal{O}_I$.

- (ii) *Conversely, let $x \in X_2$ have compact A -orbit of length $\ell > 0$, and put $\varepsilon = e^{\ell/2}$. Then ε is a real quadratic algebraic unit with conjugate ε^{-1} . If $K = \mathbb{Q}(\varepsilon)$, there is a lattice $I \subseteq K$, containing $\mathbb{Z}[\varepsilon]$, such that*

$$xA = x_I A.$$

Moreover $\varepsilon = \varepsilon_I$; equivalently, $\mathcal{O}_I^{\times,+} = \langle \varepsilon \rangle$.

Before the proof, note that taking $I = \mathcal{O}$ for an order $\mathcal{O} \subseteq K$ gives $\mathcal{O}_I = \mathcal{O}$ (an order is exactly its own multiplier ring when considered as a lattice in K), so that by Proposition 2.31 (i) *every* order in *every* real quadratic field is realised by a compact orbit. Note also that $\mathbb{Z}[\varepsilon]$ may be strictly smaller than $\mathcal{O}_I$; see Example ?? below.

Proof. For a lattice $\Lambda \subseteq \mathbb{R}^2$ write $\text{Stab}_A(\Lambda) = \{a \in A : a\Lambda = \Lambda\}$. By Section 1.2(a) and the definition (2.11) of the flow we have $\Sigma_x = \{t \in \mathbb{R} : a_t \Lambda_x = \Lambda_x\}$. Observe also that

$$(2.19) \quad \text{Stab}_A(D\Lambda) = \text{Stab}_A(\Lambda) \quad \text{for every invertible diagonal } D,$$

because diagonal matrices commute. In particular the stabiliser is unchanged by homotheties.

Step 1: for a lattice $I \subseteq K$ one has

$$\text{Stab}_A(\iota(I)) = \{m_\delta : \delta \in \mathcal{O}_I^{\times,+}\}.$$

If $\delta \in \mathcal{O}_I^{\times,+}$, then m_δ is diagonal with positive entries and $\det m_\delta = N(\delta) = 1$ by (2.17), so $m_\delta \in A$; and $m_\delta \iota(I) = \iota(\delta I) = \iota(I)$ since $\delta I = I$. Conversely let $a \in A$ with $a \iota(I) = \iota(I)$. Choose $\alpha \in I \setminus \{0\}$. Then $a \iota(\alpha) = \iota(\beta)$ for some $\beta \in I$, necessarily non-zero, and comparing the two coordinates with $\delta := \beta/\alpha \in K^\times$ gives $a = m_\delta$. Hence $\iota(\delta I) = m_\delta \iota(I) = \iota(I)$, so $\delta I = I$ by injectivity of ι ; thus $\delta \in \mathcal{O}_I^\times$, while $\delta > 0$ because the entries of a are positive and $N(\delta) = \det a = 1$.

Step 2: proof of (i). By (2.19), $\text{Stab}_A(\Lambda_{x_I}) = \text{Stab}_A(\iota(I))$, which by Step 1 and the definition of ε_I equals $\{m_{\varepsilon_I^k} : k \in \mathbb{Z}\}$. Since $N(\varepsilon_I) = 1$ we have $\bar{\varepsilon}_I = \varepsilon_I^{-1}$, so

$$m_{\varepsilon_I} = \begin{pmatrix} \varepsilon_I^{-1} & \\ & \varepsilon_I \end{pmatrix} = a_{2 \log \varepsilon_I},$$

whence $\Sigma_{x_I} = (2 \log \varepsilon_I) \mathbb{Z}$. The orbit map therefore factors through the circle $\mathbb{R}/(2 \log \varepsilon_I) \mathbb{Z}$, so $x_I A$ is compact of length $2 \log \varepsilon_I$. Finally, ε_I , and therefore the length $2 \log \varepsilon_I$, depends only on $\mathcal{O}_I$.

Step 3: proof of (ii). Write $\Lambda := \Lambda_x$. By Lemma 2.26 we have $\Sigma_x = \ell \mathbb{Z}$, so

$$a_\ell = \begin{pmatrix} \varepsilon^{-1} & \\ & \varepsilon \end{pmatrix} \in \text{Stab}_A(\Lambda), \quad \varepsilon = e^{\ell/2} > 1.$$

As computed above, ε is a root of $\lambda^2 - n\lambda + 1$ with $n = \text{tr } \gamma \in \mathbb{Z}$, $n \geq 3$; this is its minimal polynomial since $\varepsilon \notin \mathbb{Q}$, so ε is an algebraic integer, $K = \mathbb{Q}(\varepsilon)$ is real quadratic, the second root ε^{-1} is the Galois conjugate $\bar{\varepsilon}$, and $\mathbb{Z}[\varepsilon] = \mathbb{Z} \oplus \mathbb{Z}\varepsilon$ is an order of K . With the notation above this says precisely

$$a_\ell = m_\varepsilon.$$

Since $\mathbb{Z}[\varepsilon] = \mathbb{Z} + \mathbb{Z}\varepsilon$ and $m_\varepsilon \Lambda = \Lambda$, we have $m_\delta \Lambda = \Lambda$ for all $\delta \in \mathbb{Z}[\varepsilon]$. Thus Λ is naturally a $\mathbb{Z}[\varepsilon]$ -module.

Choose $v = (v_1, v_2) \in \Lambda \setminus \{0\}$. Then $v_1 v_2 \neq 0$: if, say, $v_2 = 0$, then $a_\ell^k v = (\varepsilon^{-k} v_1, 0)$ is a non-zero element of Λ for every $k \geq 0$ and tends to 0, contradicting the discreteness of Λ ; if $v_1 = 0$, the same contradiction is obtained by applying negative powers of a_ℓ . We may moreover assume $v_1 v_2 > 0$: the element $\delta := 1 - \varepsilon \in \mathbb{Z}[\varepsilon]$ has norm

$$N(\delta) = (1 - \varepsilon)(1 - \varepsilon^{-1}) = 2 - n < 0,$$

so if $v_1 v_2 < 0$ we may replace v by $m_\delta v \in \Lambda$, whose coordinates have product $N(\delta) v_1 v_2 > 0$.

Set $D := \text{diag}(v_1, v_2)$ and $\Lambda' := D^{-1} \Lambda$. Then Λ' is again a lattice, still invariant under $\mathbb{Z}[\varepsilon]$ because D commutes with diagonal matrices, and it contains $D^{-1} v = (1, 1) = \iota(1)$. Consequently

$$\iota(\mathbb{Z}[\varepsilon]) = m(\mathbb{Z}[\varepsilon]) \iota(1) \subseteq \Lambda'.$$

Both sides are lattices in $\mathbb{R}^2$, so the index $N := [\Lambda' : \iota(\mathbb{Z}[\varepsilon])]$ is finite and $N \Lambda' \subseteq \iota(\mathbb{Z}[\varepsilon])$, whence

$$\Lambda' \subseteq \frac{1}{N} \iota(\mathbb{Z}[\varepsilon]) = \iota\left(\frac{1}{N} \mathbb{Z}[\varepsilon]\right) \subseteq \iota(K).$$

Thus $I := \iota^{-1}(\Lambda')$ is a finitely generated $\mathbb{Z}$ -module containing $\mathbb{Z}[\varepsilon]$; since $K = \mathbb{Q}(\varepsilon)$, it is a lattice in K .

It remains to compare $\Lambda = D \iota(I)$ with $\iota(I)$. Since $v_1 v_2 > 0$ the two diagonal entries of D have the same sign. Put $r := (v_1 v_2)^{1/2} > 0$. The matrix $r^{-1} D$ has determinant 1, so $D = \pm r a$ with $a \in A$; since $I = -I$ we have $\iota(I) = -\iota(I)$ and hence, in either case,

$$\Lambda = r a \iota(I).$$

Comparing covolumes gives $r = \text{covol}(\iota(I))^{-1/2}$, that is $\Lambda = a \Lambda_{x_I}$, and therefore $x_A = x_I A$.

Finally, by (2.19) and Step 1,

$$\{a_t : t \in \ell \mathbb{Z}\} = \text{Stab}_A(\Lambda) = \text{Stab}_A(\iota(I)) = \{m_{\varepsilon_I^k} : k \in \mathbb{Z}\},$$

and both $a_\ell = m_\varepsilon$ and m_{ε_I} are the generators of this group with second diagonal entry > 1 ; hence $\varepsilon = \varepsilon_I$. $\square$

How about dense A -orbits?

How about equidistributed A -orbits?

2.8. The horocycle flow — the subgroup $U^- = N$. The (stable) horocycle flow on X_2 is the action of

$$U^- = \left\{ u_s = u_s^- = \begin{pmatrix} 1 & s \\ & 1 \end{pmatrix} : s \in \mathbb{R} \right\}, \quad u_s : x \mapsto u_s \cdot x = x \begin{pmatrix} 1 & -s \\ & 1 \end{pmatrix}.$$

The matrices u_s are unipotent, i.e. 1 is their only eigenvalue. Geometrically the orbits of U^- on $T^1\mathbb{H}$ are the circles in $\mathbb{H}$ tangent to $\mathbb{R}$, with the normal vectors pointing inwards, together with the horizontal lines with the vectors pointing upwards; see Figure 7.

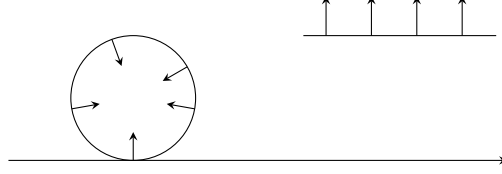


FIGURE 7. The two types of horocycle orbits in $T^1\mathbb{H}$.

The relevance of U^- for the geodesic flow is that it is precisely the *stable horospherical subgroup*:

$$(2.20) \quad U^- = \left\{ g \in \mathrm{SL}_2(\mathbb{R}) : a_t g a_t^{-1} \rightarrow I \text{ as } t \rightarrow \infty \right\}.$$

Likewise $U^+ = \left\{ u_s^+ = \begin{pmatrix} 1 & \\ s & 1 \end{pmatrix} : s \in \mathbb{R} \right\}$ is the *unstable horospherical subgroup*, obtained from U^- by letting $t \rightarrow -\infty$; it is conjugate to U^- by the Weyl element. Both claims follow from the following computation.

Remark 2.32. For all $s, t \in \mathbb{R}$,

$$a_t u_s^- a_t^{-1} = u_{e^{-t}s}^-, \quad a_t u_s^+ a_t^{-1} = u_{e^t s}^+.$$

In particular A normalises $U^\pm$ and rescales the natural parameter by $e^{\mp t}$. Indeed, with $a_t = \mathrm{diag}(\alpha, \alpha^{-1})$, $\alpha = e^{-t/2}$,

$$\begin{pmatrix} \alpha & \\ & \alpha^{-1} \end{pmatrix} \begin{pmatrix} 1 & s \\ & 1 \end{pmatrix} \begin{pmatrix} \alpha^{-1} & \\ & \alpha \end{pmatrix} = \begin{pmatrix} 1 & \alpha^2 s \\ & 1 \end{pmatrix},$$

and similarly for U^+ with α^{-2} in the lower left corner. This proves (2.20), since a matrix $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ satisfies $a_t g a_t^{-1} = \begin{pmatrix} a & e^{-t}b \\ e^t c & d \end{pmatrix} \rightarrow I$ if and only if $c = 0$ and $a = d = 1$. It also explains the contraction property

$$(2.21) \quad d_{X_2}(a_t \cdot x, a_t \cdot (u_s \cdot x)) \rightarrow 0 \quad (t \rightarrow +\infty)$$

for all $x \in X_2$ and $s \in \mathbb{R}$: indeed $a_t \cdot (u_s \cdot x) = (a_t u_s a_t^{-1}) \cdot (a_t \cdot x) = u_{e^{-t}s} \cdot (a_t \cdot x)$, and $u_{e^{-t}s} \rightarrow I$. (We will introduce the metric d_{X_2} and verify (2.21) carefully in the next section.)

The subgroup U^- also acts ergodically on (X_2, m_{X_2}) . In sharp contrast with the case of A , however, the list of possible orbits is very short (we will prove this in Chapter 5; the general statement is Ratner's orbit closure theorem):

- *Compact trajectories*, for example $\mathrm{SL}_2(\mathbb{Z})U^-$, which corresponds to the horizontal orbit through $(i, \uparrow) \in T^1\mathbb{H}$ and is periodic of period 1 because $\begin{pmatrix} 1 & 1 \\ & 1 \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$;
- *dense trajectories*, which are moreover automatically equidistributed with respect to m_{X_2} , both in the past and in the future.

This dichotomy is one more instance of the correspondence between closed orbits and rational objects, as the next section shows.

2.9. Periodic horocycles and cusps. For $g \in \mathrm{SL}_2(\mathbb{R})$ let $g \cdot \infty \in \mathbb{R} \cup \{\infty\}$ denote the image of ∞ under the extension of (2.2) to $\partial\mathbb{H} = \mathbb{R} \cup \{\infty\}$, that is $g \cdot \infty := \lim_{z \rightarrow \infty} g \cdot z$ (here $z \rightarrow \infty$ means that $|z| \rightarrow +\infty$). One checks that for $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$g \cdot \infty = \begin{cases} a/c & \text{if } c \neq 0; \\ \infty & \text{if } c = 0. \end{cases}$$

The rational boundary points $\mathbb{Q} \cup \{\infty\}$ are called *cusps*. One checks that the cusps are precisely the Γ -translates of ∞ : $\mathbb{Q} \cup \{\infty\} = \mathrm{SL}_2(\mathbb{Z}) \cdot \infty$.

Proposition 2.33. *For every $g \in \mathrm{SL}_2(\mathbb{R})$,*

$$\mathrm{SL}_2(\mathbb{Z})gU^- \text{ is compact } \iff g \cdot \infty \in \mathbb{Q} \cup \{\infty\},$$

and in that case $\mathrm{SL}_2(\mathbb{Z})gU^- = \mathrm{SL}_2(\mathbb{Z})aU^-$ for some $a \in A$.

In other words, the horocycle through $\mathrm{SL}_2(\mathbb{Z})g$ is periodic exactly when its boundary point $g \cdot \infty$ is a cusp.

Proof. Compactness implies rationality. By Lemma 2.26 (applied to $H = U^-$) there is $s \neq 0$ with $\gamma := gu_s g^{-1} \in \mathrm{SL}_2(\mathbb{Z})$. Since u_s fixes ∞ , the element γ fixes $g \cdot \infty$. Since $u_s \neq I$ is unipotent, the element γ is unipotent and non-trivial. In particular, $\mathrm{tr} \gamma = 2$. Write $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. One checks that its fixed points in $\partial\mathbb{H}$ are the solutions of $cz^2 + (d-a)z - b = 0$. If $c = 0$ the only fixed point is ∞ ; if $c \neq 0$ the discriminant is $(d-a)^2 + 4bc = (a+d)^2 - 4 = 0$ (here we used that $2 = \mathrm{tr} \gamma = a+d$), so the unique fixed point is $\frac{a-d}{2c} \in \mathbb{Q}$. Either way $g \cdot \infty \in \mathbb{Q} \cup \{\infty\}$.

Rationality implies compactness, and the second assertion. Suppose $g \cdot \infty \in \mathbb{Q} \cup \{\infty\}$ and choose $h \in \mathrm{SL}_2(\mathbb{Z})$ with $h \cdot \infty = g \cdot \infty$: for $g \cdot \infty = p/q$ in lowest terms take $h = \begin{pmatrix} p & * \\ q & * \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$, which exists since $\gcd(p, q) = 1$; for $g \cdot \infty = \infty$ take $h = I$. Then $(h^{-1}g) \cdot \infty = \infty$, which forces $h^{-1}g$ to be upper triangular, say

$$h^{-1}g = \pm \begin{pmatrix} \alpha & \beta \\ 0 & \alpha^{-1} \end{pmatrix} = \pm u_{\alpha\beta} a \quad \text{with } a = \mathrm{diag}(\alpha, \alpha^{-1}) \in A, \alpha > 0.$$

Since $-I, h \in \Gamma$ and $u_{\alpha\beta} \in U^-$, and since A normalises U^- by Remark 2.32, we obtain

$$\Gamma g U^- = \Gamma u_{\alpha\beta} a U^- = \Gamma U^- a = \Gamma a U^-.$$

Finally $\Gamma a U^-$ is compact: by Remark 2.32, $au_s a^{-1} = u_{\alpha^2 s}$, so $\Gamma a u_{\alpha^{-2}} = \Gamma u_1 a = \Gamma a$, and the orbit is the continuous image of the circle $\mathbb{R}/\alpha^{-2}\mathbb{Z}$. $\square$

2.10. The subgroups K and B . Besides A and $U^\pm$, the group $\mathrm{SL}_2(\mathbb{R})$ has two further connected subgroups of importance:

$$K = \mathrm{SO}_2(\mathbb{R}), \quad B = U^- A = \left\{ \begin{pmatrix} e^{-t/2} & s \\ 0 & e^{t/2} \end{pmatrix} : s, t \in \mathbb{R} \right\}.$$

For these two, the correspondence between closed orbits and rational objects breaks down completely, for opposite reasons.

Every K -orbit is compact, simply because K is compact and orbits are continuous images of K . (Geometrically, the K -orbit of a point of $T^1\mathbb{H}$ is the fibre of $T^1\mathbb{H} \rightarrow \mathbb{H}$, i.e. the circle of directions at a point of the modular surface.) So compactness of an orbit carries no arithmetic information at all.

At the other extreme:

Proposition 2.34. *Every B -orbit in X_2 is dense.*

Proof. Let $x \in X_2$. If the orbit xU^- is not periodic then, by the classification of U^- -orbits quoted in the previous subsection, it is dense, and so is the larger set xB .

If xU^- is periodic, of period $s_0 > 0$ say, consider the points $x_t := a_t \cdot x \in xB$ (recall $B = U^-A = AU^-$). By Remark 2.32,

$$(xa_t^{-1})u_s = x(a_t^{-1}u_s a_t)a_t^{-1} = (xu_{e^t s})a_t^{-1},$$

so $x_t U^-$ is again periodic, of period $e^{-t}s_0$ — letting $t \rightarrow -\infty$ we obtain periodic horocycle orbits inside xB of arbitrarily large period. Now closed horocycles of increasing period equidistribute in X_2 with respect to m_{X_2} (a theorem of Sarnak [S], see also Section 5.3.1), and in particular their union is dense. Hence xB is dense. $\square$

Thus a B -orbit is never closed, whatever the arithmetic nature of the point — again no information. The moral is that the phenomenon observed for A and U^- , namely

$$\text{closed orbit} \longleftrightarrow \text{rational (or arithmetic) object},$$

is a subtle one: it holds only in certain situations, and identifying them correctly is one of the themes of these notes, starting in Chapter 3.

We close the section by justifying the assertion that the above list is complete.

Remark 2.35 (The connected subgroups of $\mathrm{SL}_2(\mathbb{R})$). *Up to conjugation, the connected subgroups of $\mathrm{SL}_2(\mathbb{R})$ are*

$$\{e\}, \quad K, \quad A, \quad U^-, \quad B, \quad \mathrm{SL}_2(\mathbb{R}).$$

Since a connected subgroup is determined by its Lie subalgebra, it suffices to classify the subalgebras of $\mathfrak{sl}_2(\mathbb{R}) = \{v \in \mathrm{Mat}_2(\mathbb{R}) : \mathrm{tr} v = 0\}$ up to the adjoint action.

In dimension one, let $v \neq 0$. Its characteristic polynomial is $\lambda^2 + \det v$, so there are three cases. If $\det v < 0$, then v has two distinct real eigenvalues $\pm\lambda$ and is diagonalisable over $\mathbb{R}$; if $\det v > 0$, the eigenvalues are $\pm i\theta$ and v is conjugate over $\mathbb{R}$ to $\theta \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$; if $\det v = 0$ then $v^2 = 0$ and v is conjugate to a multiple of $\begin{pmatrix} 1 & \\ & 1 \end{pmatrix}$. In each case the conjugating matrix can be taken in $\mathrm{SL}_2(\mathbb{R})$: rescaling it does not change the conjugation, and if its determinant is negative one composes with $\begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$ resp. $\mathrm{diag}(1, -1)$, which changes the normal form only by a sign and hence does not change the line $\mathbb{R}v$. So $\mathbb{R}v$ is conjugate to the Lie algebra of A , of K or of U^- .

In dimension two, let $\mathfrak{h} \subseteq \mathfrak{sl}_2(\mathbb{R})$ be a subalgebra with $\dim \mathfrak{h} = 2$. It cannot be abelian: the centraliser in $\mathfrak{sl}_2(\mathbb{R})$ of any non-zero element is one-dimensional, as one checks on the three normal forms above. Hence $[\mathfrak{h}, \mathfrak{h}]$ is one-dimensional, say $[\mathfrak{h}, \mathfrak{h}] = \mathbb{R}v$, and we may choose $w \in \mathfrak{h}$ with $[w, v] = cv$, $c \neq 0$. Then $\mathrm{ad}(w)$ has the non-zero real eigenvalue c , which rules out w elliptic or nilpotent (whose ad -eigenvalues are $0, \pm 2i\theta$ resp. 0 only), so w is conjugate to a multiple of $\mathrm{diag}(1, -1)$; and v , being an eigenvector of $\mathrm{ad}(w)$ for a non-zero eigenvalue, is then a multiple of $\begin{pmatrix} 1 & \\ & 1 \end{pmatrix}$ or of $\begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$. In both cases $\mathfrak{h}$ is conjugate to the Lie algebra of B .

Finally $\mathfrak{sl}_2(\mathbb{R})$ is simple, so it has no subalgebra of dimension three other than itself.

bis hier.

3. THE MARGULIS BANANA TRICK

We now prove the equidistribution statement about long periodic horocycles used in §1.8. The proof is a useful model for a general principle in homogeneous dynamics: a singular measure supported on a lower-dimensional orbit can sometimes be *thickened* in directions which are contracted by the dynamics. The thickened measure is absolutely continuous with respect to Haar measure, and mixing can then be applied. This argument goes back to Margulis and is usually called the *banana trick*.⁴

Throughout this section

$$G = \mathrm{SL}_2(\mathbb{R}), \quad \Gamma = \mathrm{SL}_2(\mathbb{Z}), \quad X_2 = G/\Gamma,$$

and m_{X_2} denotes the G -invariant probability measure. We retain the notation

$$a_t = \begin{pmatrix} e^{-t/2} & 0 \\ 0 & e^{t/2} \end{pmatrix}, \quad u_s = u_s^- = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}.$$

Recall that our action convention is $g \cdot x = xg^{-1}$.

3.1. Mixing and transverse coordinates. The one dynamical input is stronger than the ergodicity mentioned in §1.3. We use it here as a black box; later it will follow from the general mixing theory for homogeneous spaces.

[Mixing of the geodesic flow] For all $f_1, f_2 \in L^2(X_2, m_{X_2})$,

$$\int_{X_2} f_1(xa_t) f_2(x) dm_{X_2}(x) \longrightarrow \left(\int_{X_2} f_1 dm_{X_2} \right) \left(\int_{X_2} f_2 dm_{X_2} \right)$$

as $|t| \rightarrow \infty$.

Since $xa_t = a_{-t} \cdot x$, this is exactly mixing for the A -action in our convention.

The closed horocycles are tangent to the U^- -direction. To thicken them we need the two complementary directions. Put

$$U^+ = \left\{ u_r^+ = \begin{pmatrix} 1 & 0 \\ r & 1 \end{pmatrix} : r \in \mathbb{R} \right\}, \quad B^+ = U^+ A.$$

Thus B^+ is the lower-triangular Borel subgroup, opposite to the subgroup $B = U^- A$ introduced in §1.8. Its Lie algebra is $\mathfrak{b}^+ = \mathfrak{u}^+ \oplus \mathfrak{a}$, and

$$\mathfrak{sl}_2(\mathbb{R}) = \mathfrak{u}^- \oplus \mathfrak{b}^+.$$

Lemma 3.1 (Local product coordinates). *The multiplication map*

$$U^- \times B^+ \longrightarrow G, \quad (u, b) \longmapsto ub,$$

is a diffeomorphism from a neighborhood of (e, e) onto a neighborhood of e in G .

Let ds be the Haar measure on U^- given by $s \mapsto u_s$, and let $m_{B^+}^{(r)}$ be a right Haar measure on B^+ . After multiplying $m_{B^+}^{(r)}$ by a constant, the fixed Haar measure m_G satisfies, on such a product neighborhood,

$$dm_G(u_s b) = ds dm_{B^+}^{(r)}(b).$$

Proof. The differential of the multiplication map at (e, e) is the linear map

$$\mathfrak{u}^- \oplus \mathfrak{b}^+ \longrightarrow \mathfrak{sl}_2(\mathbb{R}), \quad (X, Y) \longmapsto X + Y,$$

⁴The presentation in this section follows Andreas Wieser, *On the “banana”-trick of Margulis* (2018).

which is an isomorphism. The inverse function theorem gives the first claim. The second claim is the standard local product decomposition of Haar measure. The use of a *right* Haar measure on B^+ is important here, since B^+ is not unimodular. $\square$

We shall use rectangular neighborhoods in B^+ . For $\eta > 0$ small set, for instance,

$$\mathcal{O} = \{u_r^+ a_\tau : |r| < \eta, |\tau| < \eta\} \subset B^+.$$

The conjugation formulas from §1.6 give

$$a_{-t} u_r^+ a_t = u_{e^{-t}r}^+,$$

and hence

$$(3.1) \quad \mathcal{O}_t := a_{-t} \mathcal{O} a_t \subset \mathcal{O} \quad (t \geq 0).$$

In other words, when we move towards longer and longer closed horocycles, the transverse unstable direction becomes exponentially thinner, while the A -direction keeps the same size.

3.2. The family of closed horocycles. Let

$$P_t := \Gamma a_t U^- \subset X_2.$$

By the computation in §1.6,

$$a_t u_s a_t^{-1} = u_{e^{-t}s},$$

so

$$\Gamma a_t u_s = \Gamma a_t \iff e^{-t}s \in \mathbb{Z}.$$

Thus P_t is periodic of period e^t . Its normalized orbit measure is therefore

$$(3.2) \quad \mu_t(f) := e^{-t} \int_0^{e^t} f(\Gamma a_t u_s) ds, \quad f \in C_c(X_2).$$

By §1.7 every periodic U^- -orbit is of this form.

The basic difficulty is that μ_t is supported on a one-dimensional subset of the three-dimensional space X_2 , so mixing cannot be applied to (3.2) directly. We now thicken P_t in the B^+ -directions.

Choose $\mathcal{O}$ sufficiently small that the product coordinates from Lemma 3.1 are valid on $\{u_s : 0 \leq s < 1\}\mathcal{O}$. Define

$$S_0 := \{u_s b : 0 \leq s < 1, b \in \mathcal{O}\}$$

and, for $t \geq 0$,

$$(3.3) \quad S_t := \{u_s b : 0 \leq s < e^t, b \in \mathcal{O}_t\}.$$

The contraction in (3.1) is chosen precisely so that

$$(3.4) \quad S_t = a_{-t} S_0 a_t.$$

Indeed, $a_{-t} u_s a_t = u_{e^t s}$.

We shall also need that the thickening does not fold over itself in the quotient.

Lemma 3.2 (Injectivity of the initial thickening). *If $\mathcal{O}$ is sufficiently small, the quotient map $\pi : G \rightarrow G/\Gamma$ is injective on S_0 .*

Proof. Suppose

$$\Gamma u_{s_1} b_1 = \Gamma u_{s_2} b_2, \quad 0 \leq s_1, s_2 < 1, \quad b_1, b_2 \in \mathcal{O}.$$

Then, with $b = b_1 b_2^{-1}$,

$$u_{s_1} b u_{-s_2} \in \Gamma.$$

After shrinking $\mathcal{O}$, every $b \in \mathcal{O} \mathcal{O}^{-1}$ can be written as

$$b = \begin{pmatrix} \alpha & 0 \\ \beta & \alpha^{-1} \end{pmatrix}$$

with α arbitrarily close to 1 and β arbitrarily close to 0. A direct multiplication gives

$$u_{s_1} b u_{-s_2} = \begin{pmatrix} \alpha + \beta s_1 & \alpha^{-1} s_1 - \alpha s_2 - \beta s_1 s_2 \\ \beta & \alpha^{-1} - \beta s_2 \end{pmatrix}.$$

Since this matrix lies in $\mathrm{SL}_2(\mathbb{Z})$, its lower-left entry β is an integer; if $\mathcal{O}$ is small enough this forces $\beta = 0$. The two diagonal entries are then the integers α and α^{-1} , both close to 1, so $\alpha = 1$. Hence $u_{s_1-s_2} \in \mathrm{SL}_2(\mathbb{Z})$, and therefore $s_1 - s_2 \in \mathbb{Z}$. Since $s_1, s_2 \in [0, 1]$, we obtain $s_1 = s_2$ and then $b_1 = b_2$. $\square$

Let

$$\tilde{P}_0 := \pi(S_0) \subset X_2.$$

By Lemmas 3.1 and 3.2, Haar measure on S_0 pushes forward to the restriction of m_{X_2} to $\tilde{P}_0$.

3.3. Equidistribution of long closed horocycles.

Theorem 3.3 (Equidistribution of long closed horocycles). *For every $f \in C_c(X_2)$,*

$$(3.5) \quad e^{-t} \int_0^{e^t} f(\Gamma a_t u_s) ds \longrightarrow \int_{X_2} f dm_{X_2} \quad (t \rightarrow \infty).$$

Consequently, if (P_n) is any sequence of periodic U^- -orbits whose periods tend to infinity, then their normalized orbit measures converge weakly to m_{X_2} .

Proof. Fix $f \in C_c(X_2)$ and $\varepsilon > 0$. By continuity of the right G -action and compactness of the support of f , there is a neighborhood V of e in G such that

$$(3.6) \quad |f(xg) - f(x)| < \varepsilon \quad \text{for all } x \in X_2, g \in V.$$

Shrink the rectangular neighborhood $\mathcal{O}$ above so that $\mathcal{O} \subset V$, while retaining Lemma 3.2. Then $\mathcal{O}_t \subset \mathcal{O} \subset V$ for every $t \geq 0$.

Average first in the transverse directions and set

$$J_t(f) := \frac{e^{-t}}{m_{B^+}^{(r)}(\mathcal{O}_t)} \int_{\mathcal{O}_t} \int_0^{e^t} f(\Gamma a_t u_s b) ds dm_{B^+}^{(r)}(b).$$

By (3.6),

$$(3.7) \quad |\mu_t(f) - J_t(f)| < \varepsilon \quad (t \geq 0).$$

By the product decomposition of Haar measure from Lemma 3.1,

$$J_t(f) = \frac{1}{m_G(S_t)} \int_{S_t} f(\Gamma a_t g) dm_G(g).$$

The group $G = \mathrm{SL}_2(\mathbb{R})$ is unimodular, so Haar measure is invariant under inner automorphisms. Using (3.4) and the substitution $g = a_{-t}ha_t$ gives

$$(3.8) \quad J_t(f) = \frac{1}{m_G(S_0)} \int_{S_0} f(\Gamma ha_t) dm_G(h).$$

Thus the moving, increasingly thin neighborhood of the long horocycle has become a *fixed* positive-volume set translated by a_t .

By injectivity of π on S_0 , the last expression equals

$$\frac{1}{m_{X_2}(\tilde{P}_0)} \int_{\tilde{P}_0} f(xa_t) dm_{X_2}(x).$$

Put

$$F_0 := \frac{\mathbf{1}_{\tilde{P}_0}}{m_{X_2}(\tilde{P}_0)} \in L^2(X_2).$$

Then

$$J_t(f) = \int_{X_2} f(xa_t) F_0(x) dm_{X_2}(x).$$

Since $\int F_0 dm_{X_2} = 1$, mixing (Theorem 3.1) yields

$$J_t(f) \longrightarrow \int_{X_2} f dm_{X_2}.$$

Together with (3.7), and since $\varepsilon > 0$ was arbitrary, this proves (3.5).

For the final assertion, §1.7 shows that every periodic U^- -orbit is $\Gamma a_t U^-$ for some t , and its period is e^t . Thus periods tending to infinity are exactly the case $t \rightarrow \infty$ already treated. $\square$

Remark 3.4 (Why a banana?). *At time 0 the set S_0 is an ordinary small product neighborhood of one period of the closed horocycle. To compare with the orbit of period e^t we conjugate this neighborhood by a_{-t} and a_t . The unstable width is then multiplied by e^{-t} , while the horocycle direction is stretched by e^t . In geometric coordinates the resulting long, thin neighborhood bends around the closed horocycle, producing the shape that gave the trick its name. The essential point is not the picture but the identity $S_t = a_{-t}S_0a_t$: after thickening, the desired average becomes a matrix coefficient of the geodesic flow, and mixing does the rest.*

Remark 3.5 (A second application). *The same mechanism proves equidistribution of large hyperbolic circles. One starts from a compact $K = \mathrm{SO}_2(\mathbb{R})$ -orbit, thickens it in an appropriate contracted $U^\pm A$ -direction, and then applies mixing to the resulting positive-volume set. Thus the banana trick is not specific to horocycles: it is a general device for converting the translate of a lower-dimensional homogeneous measure into an object to which mixing applies.*

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